

ERGONOMIC DESIGN AND THERMAL EVALUATION OF PHASE CHANGE MATERIAL BASED COOLING VEST FOR OUTDOOR WORKERS

A Project Report

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University in partial fulfillment of requirements for
the award of degree*

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in

Industrial Engineering

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CERTIFICATE

This is to certify that this project entitled **ERGONOMIC DESIGN & THERMAL EVALUATION OF PHASE CHANGE MATERIAL BASED COOLING VEST FOR OUTDOOR WORKERS** submitted by **Toby S (TVE22IE066), Sai Krishnan K (TVE22IE058), Aravind R Krishna (LTVE22IE070) & Mohammad Basith A (LTVE22IE075)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Industrial Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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We undersigned hereby declare that the project report **Ergonomic Design & Thermal Evaluation of Phase Change Material Based Cooling Vest for Outdoor Workers**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Dr. Binoosh S.A**, Professor Mechanical Engineering

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Abstract

Outdoor and industrial workers such as traffic police, delivery riders, and construction personnel are frequently exposed to high temperatures and intense sunlight for prolonged durations. Continuous exposure to such environments results in heat stress, fatigue, dehydration, and reduced work efficiency, posing significant risks to occupational health and safety. In this context, wearable cooling technologies using Phase Change Materials (PCMs) have emerged as an effective and sustainable solution for maintaining thermal comfort.

The primary aim of this project is to design, develop, and evaluate an ergonomic PCM-based cooling vest capable of reducing heat stress while ensuring comfort, flexibility, and ease of movement for outdoor workers. The vest is designed using SolidWorks to model PCM placement and garment structure based on anthropometric data. Ansys Fluent simulations are performed to analyse heat transfer, PCM melting behaviour, and thermal efficiency under varying ambient conditions.

Experimental testing and simulation studies can be determined that the PCM vest will maintain skin temperature between 30–34°C, significantly reducing fatigue and discomfort compared to normal clothing. The results can be determined as the vest provides effective, passive, and reusable cooling without restricting movement. The developed PCM-based cooling vest offers a cost-effective and ergonomic solution for mitigating heat stress, improving worker safety, and enhancing overall productivity in high-temperature occupational environments.

Keywords - Thermal regulation, Human comfort modelling, Occupational health, Simulation analysis, Sustainable wearable technology.

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CHAPTER 1

INTRODUCTION

Outdoor and industrial workers such as traffic police, road construction labourers, delivery personnel, and MGNREGA employees are frequently exposed to extreme environmental conditions, particularly high temperatures and direct sunlight. Prolonged exposure to such heat stress environments leads to physical fatigue, dehydration, loss of concentration, and, in severe cases, heat-related illnesses. These issues not only endanger the health and safety of workers but also significantly reduce productivity and work efficiency.

Traditional cooling solutions—such as electric fans, air-conditioned rest areas, and ice-based cooling packs—offer only temporary relief and are often impractical for workers engaged in mobile or field-based tasks. Therefore, there is a growing demand for a portable, efficient, and ergonomic solution to help regulate body temperature and maintain comfort throughout the workday.

Phase Change Materials (PCMs) have emerged as an effective passive cooling technology capable of absorbing and releasing heat energy during phase transitions. When incorporated into wearable garments, such as cooling vests, PCMs can maintain a stable microclimate around the human body by keeping temperatures within a thermally comfortable range (approximately 28–32°C). Paraffin wax is one of the most widely used PCMs due to its suitable melting temperature, high latent heat capacity, chemical stability, and reusability.

The current project aims to design and evaluate an ergonomic PCM-based multilayer cooling vest that provides enhanced thermal comfort for outdoor workers operating under high-heat conditions. The vest integrates multiple functional layers—an inner layer in contact with the skin for comfort, a middle PCM layer for heat absorption and storage, and an outer protective layer for durability and insulation. The design focuses on maintaining lightweight structure, flexibility, and optimal weight distribution, ensuring it does not restrict the wearer's movement during physically demanding task

1.1 General Background

In tropical and industrial environments, workers are frequently exposed to high temperatures and direct sunlight for prolonged periods. Professions such as road construction workers, traffic police, delivery personnel, and MGNREGA (rural employment) laborers face extreme environmental heat stress that significantly impacts their health, safety, and work performance. Continuous exposure to such conditions often results in fatigue, dehydration, reduced productivity, heat exhaustion, and in severe cases, heat stroke.

Traditional cooling methods like fans, air-conditioning, or wet garments are ineffective in open outdoor conditions and cannot provide long-duration, portable cooling. As a result, there is an urgent need for personalized, passive, and energy-efficient cooling solutions for workers who cannot access conventional climate control systems.

Phase Change Materials (PCMs) have emerged as one of the most promising technologies for thermal management applications. PCMs have the unique ability to absorb, store, and release latent heat during the phase transition process (typically from solid to liquid and vice versa). When integrated into wearable garments such as cooling vests, PCMs can help maintain a comfortable body temperature by absorbing excess body heat during work in hot environments.

Among various PCMs, paraffin wax stands out due to its appropriate melting point (30–34°C), high latent heat capacity, chemical stability, and non-toxicity, making it ideal for personal cooling applications. By embedding PCM into a multilayer vest structure—comprising an inner comfort layer, a PCM middle layer, and an outer protective layer—effective heat regulation can be achieved without restricting body movement or adding significant weight.

Thus, the integration of PCM-based cooling vests represents an ergonomic and sustainable approach to reducing heat stress, improving comfort, and enhancing occupational safety

1.2 Problem Definition

Heat stress is a major occupational hazard for individuals working in **hot and humid environments**, particularly in developing countries. Outdoor workers experience fluctuating thermal conditions that can elevate core body temperature, resulting in discomfort and decreased efficiency. The absence of suitable protective garments and limited awareness of thermal comfort measures exacerbate this issue.

Existing cooling solutions, such as **ice-pack vests or fan-assisted jackets**, often have **limited duration, added weight, and restricted mobility**, making them impractical for long work hours. Moreover, these systems require external power sources or frequent recharging, which are not feasible in rural or field environments.

The main problems addressed in this study are:

- Lack of **effective, portable, and ergonomic cooling solutions** for outdoor workers.
- Insufficient data on **thermal comfort parameters and heat exposure levels** among various worker groups.
- Limited **design optimization and simulation-based evaluation** of PCM-integrated wearable systems.

Therefore, there is a pressing need to design and evaluate a **PCM-based cooling vest** that can provide sustained cooling comfort, reduce body heat load, and improve ergonomic performance during field activity.

1.3 Motivation

The motivation behind this project stems from the growing concern over occupational heat stress and its impact on human performance, health, and well-being in outdoor professions. In rural and industrial sectors, workers operate in high-temperature environments without sufficient protective gear. Continuous exposure to heat not only reduces productivity but also causes long-term health issues such as dehydration, muscle fatigue, and cardiovascular strain. The team was motivated by the potential of Phase Change Materials (PCMs) to provide a sustainable, energy-efficient, and portable cooling mechanism that can be integrated into

wearable clothing. PCM-based cooling vests are environmentally friendly, reusable, and require no external power, making them ideal for outdoor applications.

Additionally, the availability of advanced simulation software tools such as SolidWorks, ANSYS and FLUENT, enables accurate thermal and ergonomic analysis during the design process. These tools help visualize real-time heat transfer, air flow, and human comfort, reducing prototyping costs and improving design efficiency.

Ultimately, the motivation is to develop a practical solution that improves thermal comfort, occupational safety, and productivity for workers exposed to extreme heat — contributing both to worker welfare and sustainable industrial development

1.4 Objectives

The main goal of this project is to design and evaluate a PCM-based ergonomic cooling vest that reduces heat stress and enhances comfort for outdoor workers.

Specific Objectives:

1. To study and analyze the effects of heat stress and thermal discomfort among various outdoor worker groups (road workers, rural employees, traffic police, and delivery personnel).
2. To design an ergonomic multi-layer cooling vest using SolidWorks, ensuring flexibility, comfort, and weight balance.
3. To optimize the distribution, size, and placement of PCM pockets for uniform cooling across the human torso.
4. To perform transient thermal analysis using ANSYS Fluent to study temperature distribution and cooling behavior.
5. To develop ergonomic design with balanced weight distribution and flexibility for user comfort.
6. To collect and analyze field data through surveys among workers to understand real environmental conditions and validate simulation results.
7. To compare results of simulated cooling performance with field feedback for practical recommendations and improvements.
8. To prepare a complete evaluation report and presentation demonstrating design outcomes, ergonomic improvements, and feasibility

1.5 Scope

The scope of this project extends to the conceptualization, design, simulation, and evaluation of an ergonomic Phase Change Material (PCM)-based cooling vest intended for outdoor and industrial workers operating under high-temperature conditions. The study integrates multiple disciplines, including thermal engineering, ergonomics, and computational modeling, to develop a scientifically validated, user-centered design.

The scope focuses primarily on the use of paraffin wax as the PCM due to its favorable melting temperature range (22–28°C), high latent heat capacity, non-toxic nature, and cost effectiveness. The project emphasizes the development of a passive cooling garment capable of providing comfort without relying on external power sources.

The study limits itself to simulation-based design validation and survey-based evaluation of comfort, without physical prototyping or experimental testing, due to time and resource constraints. However, the outcomes of this research lay a strong foundation for future experimental studies and field validation.

The scope also includes **field data collection** through structured surveys and interviews with different categories of outdoor workers to understand their work conditions, exposure duration, hydration habits, and thermal discomfort experiences. This primary data supports the design by defining user-specific requirements and validating the potential benefits of the proposed cooling vest. The project thereby combines **ergonomic user data** and **thermal engineering principles** to produce a more realistic and human-centered solution.

However, certain aspects lie **beyond the scope** of this project. The current study does not involve laboratory testing, physiological monitoring, or real-time body temperature measurement using human participants. It also excludes detailed cost analysis, large-scale manufacturing feasibility, and long-term wear-and-tear evaluation of the material. Despite these limitations, the computational and analytical findings derived from this work will provide valuable insights for future prototype development and experimental validation.

In essence, the scope of this project is defined within the framework of a **design-simulation evaluation cycle**. It integrates theoretical concepts of phase change cooling, digital design

techniques, and ergonomic analysis to present a **feasible, lightweight, and sustainable cooling solution** for workers exposed to hot and humid conditions. The outcome is expected

to establish a foundation for further research in **smart textile design, occupational thermal comfort, and passive wearable cooling systems**, ultimately contributing to enhanced worker safety and productivity.

1.6 Research Methodology

The research methodology adopted in this study follows a systematic, multi-stage approach that integrates theoretical research, computational modeling, ergonomic design, and field based data analysis. The methodology ensures that the proposed cooling vest design is scientifically valid, ergonomically optimized, and practically applicable for outdoor workers operating in hot and humid environments.

The project began with an extensive literature review to understand the current developments in the field of thermal comfort, phase change materials (PCMs), and wearable cooling systems. Reputed databases such as *MDPI*, *Elsevier*, *IEEE Xplore*, and *Google Scholar* were referred to identify relevant studies related to PCM selection, heat transfer modeling, ergonomic apparel design, and human thermoregulation. The review helped identify suitable PCM candidates such as paraffin wax, which exhibits an optimal melting range of 22–28°C, high latent heat capacity, chemical stability, and non-toxicity—making it ideal for wearable applications. Furthermore, it revealed existing limitations in the ergonomic design of cooling vests, particularly the lack of integration between thermal performance simulation and comfort evaluation.

Following the literature study, a field data collection survey was conducted among outdoor and industrial workers, including road construction laborers, MGNREGA rural employees, traffic police officers, and delivery personnel. The survey was administered through Google Forms and direct interviews to collect data related to working hours, exposure duration, environmental temperature, hydration frequency, and perceived heat discomfort. The responses helped in identifying the major factors contributing to heat stress, user preferences in terms of clothing, and acceptable weight or size for wearable cooling garments. The survey data also provided essential insights into designing a vest that would be comfortable, lightweight, and non-

restrictive during work activities.

Based on the findings, a conceptual design of the PCM-based cooling vest was developed using SolidWorks software. The vest was structured into three functional layers: an inner layer made of bio-based, breathable fabric for comfort and moisture absorption; a middle layer containing encapsulated paraffin wax PCM cells responsible for heat absorption and gradual release; and an **outer protective layer** made of durable insulating fabric to resist external heat and mechanical stress. The SolidWorks model was optimized for **fit, flexibility, and uniform weight distribution** to ensure that the design conformed to ergonomic principles.

Once the 3D model was finalized, it was imported into ANSYS Workbench for thermal simulation and finite element analysis (FEA). The simulation aimed to evaluate the temperature distribution, heat flux, and cooling duration across the PCM layer under variable ambient temperatures representative of outdoor working conditions. The study also employed ANSYS FLUENT (Computational Fluid Dynamics) to simulate airflow and convective heat transfer around the vest, assessing how environmental air movement affects cooling performance. These analyses provided quantitative insights into the thermal response of the vest and helped in refining material thickness and PCM volume for optimal performance.

After simulations, the data analysis and validation phase combined both computational and field data. The thermal performance results from ANSYS and FLUENT were compared against environmental exposure data from worker surveys to assess real-world feasibility. The analysis focused on key parameters such as cooling time, surface temperature reduction, and user comfort index. The results confirmed that the paraffin-based PCM vest could maintain a comfortable temperature range for several hours, thereby reducing the physiological heat burden on outdoor workers.

The final phase involved documentation, interpretation, and presentation of findings. All design processes, analyses, and results were compiled into a comprehensive project report supported by relevant charts, graphs, and simulation images. A PowerPoint presentation was also developed summarizing the objectives, methodology, simulation outputs, ergonomic results, and recommendations for practical application.

The adopted methodology is thus a hybrid approach combining computational analysis, ergonomic design validation, and user-based feedback, which collectively ensures the reliability and usability of the proposed system.

CHAPTER 2

LITERATURE REVIEW

Overview

The literature on personal cooling systems and phase change material (PCM)-based wearable garments has expanded significantly in recent years due to growing awareness of occupational heat stress and the need for sustainable, energy-efficient solutions. Previous research has focused on developing materials and garments that can reduce heat accumulation, maintain thermal comfort, and enhance human performance in hot environments. However, most existing studies primarily emphasize material performance or laboratory-based evaluations, while limited work has been done to integrate ergonomic design principles and simulation based validation into the development of wearable cooling systems.

The reviewed studies consistently highlight that heat stress is one of the most critical challenges faced by industrial and outdoor workers. Prolonged exposure to high temperatures can lead to severe physiological strain, dehydration, fatigue, and productivity loss. Traditional cooling approaches—such as air-conditioned rest zones, fans, or ice-based garments—provide only temporary or localized relief and are impractical in outdoor or mobile settings. This gap has led researchers to explore Phase Change Materials (PCMs) as a passive, power-free cooling mechanism suitable for wearable applications. PCMs, particularly paraffin wax, salt hydrates, and fatty acid blends, possess the ability to absorb and release large amounts of latent heat during phase transitions, making them ideal for maintaining stable microclimate temperatures within clothing systems.

Several studies have demonstrated that PCM-integrated vests or fabrics can significantly reduce body and skin temperature under heat exposure. For instance, research published in *Nanomaterials* (MDPI, 2024) revealed that radiative and PCM cooling layers combined could lower surface temperature by 3–5°C and extend thermal comfort for over two hours. Similarly, field studies involving construction and road workers confirmed that PCM vests improved

endurance and lowered core body temperature by nearly 2°C. These studies collectively validate the thermal effectiveness of PCM technology. However, many of them lack comprehensive evaluation of comfort, mobility, and usability, which are essential parameters for workers performing dynamic physical activities.

2.2 Literature Survey

2.2.1 Vests with Radiative Cooling Materials to Improve Thermal Comfort of Outdoor Workers: An Experimental Study (Nanomaterials, MDPI, 2024)

This recent study aimed to investigate the effectiveness of radiative cooling materials integrated with wearable vests to enhance thermal comfort for outdoor workers exposed to high solar radiation. The researchers designed a specialized vest incorporating infrared-emissive nanomaterials and PCM layers to reflect solar heat while dissipating body-generated heat through radiation. The experimental trials were performed under controlled and outdoor conditions using thermal manikins and human participants.

The findings revealed that the vest successfully reduced surface temperature by 3–5°C compared to conventional fabrics and extended thermal comfort duration by approximately 2.5 hours. The combination of radiative cooling and PCM integration showed promising results in maintaining skin temperature stability and delaying the onset of heat stress. However, the research gap identified was that the vest's ergonomic comfort, movement restriction, and longterm usability were not comprehensively evaluated. This highlights the need for ergonomically optimized multilayer PCM vests that ensure both cooling performance and user comfort—an aspect addressed in the current study.

2.2.2 Comfort and Performance Improvement through the Use of Cooling Vests for Construction Workers (International Journal of Construction Safety and Technology, 2021)

This study focused on the practical performance and comfort benefits of PCM-based cooling vests among construction workers operating under direct sunlight. The authors conducted field experiments at construction sites to measure physiological parameters such as skin

temperature, heart rate, and perceived exertion when using PCM vests. The methodology involved comparing groups wearing PCM-integrated vests with those wearing standard cotton workwear during 6-hour shifts.

The results indicated that workers wearing PCM vests experienced an average reduction of 1.8°C in core body temperature and reported lower fatigue levels compared to the control group. The vests maintained thermal comfort for about 3 hours under moderate work intensity. The study concluded that PCM cooling vests enhance endurance, reduce discomfort, and improve work performance. However, the research did not include computational simulations or ergonomic analysis, limiting its application to generalized observations. The current project aims to address this gap by combining field data collection with simulation-based validation to evaluate both thermal and ergonomic performance.

2.2.3 Experimental Investigation into the Thermal Performance of Personal Cooling Mechanisms (Applied Thermal Engineering, 2020)

This experimental work compared multiple personal cooling technologies, including PCM vests, evaporative cooling garments, and fan-assisted jackets, under varying thermal conditions. The researchers employed infrared thermography and thermal sensors to analyze temperature distribution and heat dissipation rates from each system. PCM-based systems used paraffin wax and salt hydrates with different encapsulation methods.

The study found that PCM-based vests maintained lower surface temperatures for a longer duration (up to 4 hours) compared to evaporative systems, which were more dependent on humidity levels. PCM materials with melting points between 24°C and 28°C offered the best balance between comfort and phase change efficiency. The researchers also observed that PCM thickness and distribution significantly affect cooling uniformity. However, the study's limitation was its lack of ergonomic assessment and simulation-based validation of thermal flow inside the garment. analysis using ANSYS FLUENT to model internal heat transfer and validate design efficiency before physical testing.

2.2.4 Smart Firefighters' PPE: Impact of Phase Change Materials on Thermal Stress and Comfort (Safety Science, 2019)

This research explored the integration of PCMs into firefighters' personal protective equipment (PPE) to improve thermal management during high-intensity tasks. Firefighters' protective gear typically causes significant heat buildup due to its low breathability. The study incorporated PCM pouches made from n-eicosane and paraffin blends into inner protective layers of firefighting suits and conducted simulations and laboratory tests using thermal manikins.

The results demonstrated a 15–20% reduction in heat stress indicators, including mean skin temperature and sweat rate, during simulated fire exposure. The PCM-enhanced PPE significantly delayed the onset of thermal discomfort while maintaining safety standards. The study also highlighted the importance of PCM layer thickness and placement for maximizing heat absorption efficiency. However, the research primarily focused on protective clothing for emergency scenarios and not on ergonomic flexibility or comfort for prolonged daily use. The present study extends this research direction by applying PCM integration to lightweight, ergonomic vests suitable for everyday outdoor occupations rather than specialized emergency services.

2.2.5 Analytical Model for Thermoregulation of the Human Body in Contact with a PCM Cooling Vest (Journal for Future Apparel and Textile, MDPI, 2022)

This paper developed a mathematical model for predicting human thermoregulation when in contact with PCM-based cooling garments. The model incorporated bioheat transfer equations and PCM melting/solidification dynamics to simulate the thermal response of the human body under varying environmental conditions. The authors validated their analytical model through small-scale experiments using human subjects and PCM packs made of paraffin wax and fatty acid mixtures.

The study found that the PCM vest maintained a steady skin temperature of 32–34°C for approximately 3 hours under ambient temperatures of 35°C, significantly reducing physiological strain. The research also emphasized that PCM location and encapsulation quality greatly influenced the vest's effectiveness. Although the model successfully simulated thermal behavior, it lacked integration with ergonomic design parameters and posture analysis

tools. The present project addresses this research gap by combining human modeling in Siemens Jack with thermal simulation in ANSYS, enabling a holistic evaluation of both comfort and heat management.

2.2.6 Role of Phase Change Materials in Thermal Comfort Garments for BSF Jawans in Rajasthan and Kashmir

The study by Paul Gregory and Durkesh Karthik focuses on the application of Phase Change Materials (PCM) in thermal comfort garments for personnel working under extreme environmental conditions. The research highlights the importance of PCM as a thermal energy storage medium, which utilizes latent heat during phase change to regulate temperature and maintain thermal comfort.

A key contribution of the study is the selection of PCM based on environmental temperature conditions. The authors highlight that the melting temperature of PCM must be chosen according to the required thermal comfort range. For hot regions like Rajasthan, PCM with a melting temperature around 25°C is recommended so that it absorbs excess body heat when the temperature rises. Conversely, for cold regions like Kashmir, PCM with a lower melting point is selected to release heat and keep the body warm. This demonstrates that PCM selection is highly application-specific and directly influences the performance of the cooling garment. The paper also provides a detailed classification of PCM into organic, inorganic, and eutectic materials. Organic PCMs such as paraffin and polyethylene glycol are preferred for wearable applications due to their chemical stability and suitable melting ranges. Inorganic PCMs, including hydrated salts, offer higher thermal conductivity and latent heat but may face issues such as phase segregation and corrosion. Eutectic PCMs provide a combination of properties but require careful formulation.

Another important aspect discussed in the study is the selection criteria for PCM, which includes latent heat capacity, melting temperature, compatibility with garment materials, and safety. The authors identify polyethylene glycol as a suitable PCM for thermal comfort garments in hot conditions due to its favorable thermal properties and compatibility with textile materials.

To evaluate the performance of PCM-based garments, the study employs Computational Fluid Dynamics (CFD) analysis using ANSYS Fluent. A multilayer garment model is developed, and

simulations are conducted to observe temperature distribution and phase change behavior. The results show that before reaching the melting point, the PCM temperature increases along with the garment temperature. Once the melting point is reached, the PCM absorbs heat and maintains a nearly constant temperature, demonstrating the latent heat storage effect.

The simulation results also indicate that even when the ambient temperature exceeds the melting point of the PCM, the material continues to absorb heat and maintain thermal stability for a certain duration. This confirms the effectiveness of PCM in delaying temperature rise and providing sustained cooling or heating depending on the application.

2.2.7 The impact of a phase-change cooling vest on heat strain and the effect of different cooling pack melting temperatures. *European Journal of Applied Physiology*, 113(5), (2013)

Heat stress is a significant challenge in occupational environments such as firefighting, military operations, and industrial workplaces, where workers are exposed to high temperatures while wearing protective clothing that restricts heat dissipation. This leads to a condition known as uncompensable heat strain, which negatively affects human performance, safety, and endurance. To address this issue, personal cooling technologies have been developed, among which phase change material (PCM)-based cooling vests have gained considerable attention. These vests are widely preferred due to their passive operation, portability, and ability to provide effective cooling without requiring external power sources, making them suitable for real-time field applications.

A study conducted by House et al. (2013) investigated the effectiveness of PCM cooling vests with different melting temperatures (0°C, 10°C, 20°C, and 30°C) under simulated high-temperature working conditions. The results showed that during physical activity, vests with lower melting temperatures (0°C and 10°C) were significantly more effective in reducing heat strain compared to those with higher melting temperatures. This is because the larger temperature difference enhances heat absorption from the body. However, during the recovery phase, all PCM vests provided noticeable cooling, indicating that such systems are particularly beneficial during rest periods. The study also demonstrated that the use of PCM cooling vests reduces core body temperature, heart rate, and overall physiological strain, thereby improving thermal comfort and work efficiency.

Despite the higher cooling performance of lower temperature PCM, user comfort plays a crucial role in practical applications. The study found that a vest with a melting temperature of 10°C was most preferred by users, as it provided an optimal balance between effective cooling and comfort. In contrast, the 0°C vest, although more efficient in heat removal, caused discomfort and temporary skin redness due to excessive cold exposure. Higher temperature PCM vests were more comfortable but less effective in reducing heat stress. Overall, the literature suggests that PCM cooling vests are an effective solution for managing heat strain, with a melting temperature around 10°C offering the best compromise between cooling efficiency and wearer comfort.

2.2.8 Designing, Manufacturing, and Evaluating the Performance of Active Liquid Cooling Vest in Hot Laboratory Conditions (*International Journal of Environmental Health Engineering*, 11, 10).

In addition to passive PCM-based systems, active cooling technologies have also been explored to reduce heat stress. Dehghani and Dehghan (2022) conducted a study to design, manufacture, and evaluate an active liquid cooling vest (ALCV) under hot laboratory conditions. The study focused on developing a cost-effective and efficient cooling system using circulating cold water through silicone tubes embedded within the vest. The experimental evaluation was carried out in a controlled environment with high temperature and humidity, where participants performed physical activity both with and without the cooling vest. Various physiological and perceptual parameters such as heart rate, skin temperature, oral temperature, thermal sensation, and perceived exertion were measured. The results demonstrated that the active liquid cooling vest significantly reduced skin temperature, particularly in the chest and scapular regions, with a reduction of approximately 2°C compared to the non-cooling condition. The cooling vest also improved thermal sensation and reduced perceived exertion, indicating enhanced comfort and reduced physiological stress. However, the effect on core body temperature and heart rate was relatively less significant compared to skin temperature changes. This suggests that while active cooling systems are effective in improving local cooling and comfort, their impact on overall body temperature may depend on factors such as duration of use and cooling capacity.

The study further highlighted the advantages of active cooling systems over PCM-based systems, including better control over cooling intensity through adjustable flow rates and improved user mobility. Unlike PCM vests, which require pre-cooling and have limited

cooling duration, active liquid cooling vests can provide continuous cooling as long as the system is operational. However, these systems are generally more complex, require power sources, and may involve higher initial costs. The research emphasized the importance of ergonomic design, lightweight construction, and efficient heat transfer mechanisms to improve the practicality of active cooling garments. Overall, the study concluded that active liquid cooling vests are a promising solution for reducing heat stress, particularly in controlled environments where continuous cooling is required.

2.3 Research Gap

A thorough analysis of the existing literature on PCM-based cooling garments and thermal comfort systems reveals several critical research gaps that justify the need for the present study. While numerous studies have validated the thermal performance of Phase Change Materials (PCMs) in reducing body heat and enhancing comfort, there remains a significant lack of integrated research combining thermal analysis, ergonomic design, and user-centered validation.

Most of the reviewed works, such as those published in *Nanomaterials* (2024) and the *International Journal of Construction Safety and Technology* (2021), primarily focused on thermal cooling performance under experimental or controlled conditions. These studies successfully demonstrated that PCM-integrated vests could lower body and surface temperatures and prolong comfort duration. However, they did not incorporate ergonomic evaluation methods to assess how the vest interacts with the wearer's body during dynamic movement, prolonged activity, or repetitive physical tasks. In real occupational environments, factors like vest weight, flexibility, posture comfort, and fit are as important as cooling efficiency. The absence of ergonomic validation creates a gap between laboratory performance and practical usability in the field.

Another notable gap lies in the lack of simulation-based design optimization. Many studies used physical prototypes and thermal manikin testing but did not employ computational tools such as ANSYS, FLUENT, or SolidWorks for virtual evaluation. Computational modeling offers the advantage of predicting thermal gradients, heat flux distribution, and air circulation patterns across the garment layers. The absence of such simulation-driven design limits the accuracy and repeatability of experimental findings. Few studies, such as the analytical model presented in the *Journal for Future Apparel and Textile* (2022), introduced mathematical simulations of thermoregulation, but these models did not consider ergonomic factors like motion, comfort, and mobility. The integration of Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) in garment design is still relatively unexplored in existing PCM research.

Furthermore, while studies like *Experimental Investigation into the Thermal Performance of Personal Cooling Mechanisms* (2020) and *Smart Firefighters' PPE: Impact of Phase Change Materials* (2019) focused on high-intensity thermal protection, they were developed primarily for emergency responders and industrial environments, not for outdoor or rural workers engaged in long-duration physical activities. These occupational groups—such as road workers, MGNREGA employees, traffic police, and delivery personnel—represent a large population exposed to chronic heat stress but have received limited research attention. The socio-environmental context of such workers, including limited access to electricity, hydration, and rest facilities, necessitates portable, passive, and ergonomic cooling solutions, which are rarely addressed in the existing literature.

Additionally, past studies often focused narrowly on material characterization—analyzing thermal conductivity, latent heat, or melting temperature—without evaluating system-level performance in wearable configurations. For example, research on PCM-coated fabrics (*Journal of Industrial Textiles*, 2018) verified material durability and stability through multiple heating-cooling cycles but failed to assess how those materials perform when assembled into multi-layer garments. This disconnect between material-level testing and garment-level application represents a major gap in PCM apparel research.

Another important research limitation identified is the absence of real user feedback or fieldbased data. Very few studies collected information directly from workers or end-users regarding comfort preferences, acceptable weight, and design requirements. Without incorporating field data and ergonomic surveys, the design process remains largely theoretical and may not align with user expectations or practical challenges. The current project addresses

this limitation through primary data collection from various outdoor worker groups to understand actual thermal discomfort and work conditions.

Moreover, the integration of sustainability and bio-based materials in cooling garment design has not been extensively studied. Most PCMs used in earlier research rely on synthetic materials and encapsulation polymers, which may not be environmentally friendly or costeffective for large-scale production. This presents another research opportunity—to investigate bio-based or biodegradable alternatives that can complement paraffin wax while ensuring longterm sustainability and affordability.

CHAPTER 3

METHODOLOGY

The methodology adopted for the development of the Phase Change Material (PCM) based cooling vest involves a systematic approach integrating ergonomic design principles, material selection, computer-aided design (CAD) modeling, and user-centered evaluation. The objective is to design and assess a vest that provides thermal comfort and reduces heat stress for outdoor workers such as traffic police, delivery riders, and MGNREGA laborers exposed to high-temperature environments.

The project follows a **hybrid approach** combining experimental, analytical, and simulation-based techniques. The study progresses through the following major phases:

1. Preliminary Study & Literature Review
2. Identification of Research Gaps and Problem Definition
3. Data Collection and Questionnaire Survey
4. Ergonomic Evaluation and Design Optimization
5. 3D Modelling using SolidWorks
6. Thermal Analysis and Performance Evaluation
7. Obtained Results

3.1: Preliminary Study & Literature Review

The initial stage involved a comprehensive literature survey aimed at understanding the current state-of-the-art in cooling technologies, phase change materials (PCMs), and ergonomic considerations in protective clothing. Journals such as *Applied Sciences*, *Polymers*, *Thermo*, and *International Journal of Clothing Science and Technology* (2021–2025) were thoroughly reviewed. The survey emphasized:

- PCM Thermal Properties: Investigation into various PCMs, focusing on melting points, latent heat of fusion, thermal conductivity, and phase stability over repeated cycles.

- **Cooling Duration and Reusability:** Analysis of how long PCMs can sustain cooling before recharging and their ability to undergo multiple phase change cycles without significant degradation.
- **User Comfort and Ergonomics:** Assessment of clothing weight, flexibility, breathability, and mobility to ensure minimal interference with daily tasks.
- **Integration with PPE:** Evaluation of existing strategies for embedding PCMs into protective clothing while maintaining safety and usability standards.

The findings from this phase provided foundational knowledge, guiding the selection of appropriate PCMs and identifying gaps in the ergonomic design of cooling garments.

3.2: Problem Definition and Objectives:

Based on literature insights and preliminary field observations, the primary problem was defined:

“To design a lightweight, reusable, and ergonomically optimized PCM-based cooling vest that maintains comfort temperature (28–32°C) for outdoor workers and reduces heat-related stress.”

Objectives of the Project:

1. To design and develop a PCM-based cooling vest optimized for ergonomic comfort and ease of use.
2. To select suitable PCM materials ensuring effective cooling, durability, and safety.
3. To simulate and evaluate the vest’s heat transfer and ergonomic performance using SolidWorks.
4. To assess user perception, comfort, and feasibility through field surveys.

These objectives ensured a holistic approach, addressing thermal performance, usability, and user acceptance simultaneously.

3.3: Material Selection:

Material selection was critical to achieving both thermal efficiency and ergonomic comfort. The parameters considered included thermal properties, physical characteristics, safety, and compatibility with textiles.

PCM Options Considered:

- **Paraffin wax:** High latent heat, chemical stability, and wide availability.
- **Bio-based PCM:** Sustainable, non-toxic, and environmentally friendly.
- **Hydrated salts:** High energy storage capacity and cost-effectiveness.

Selection Criteria:

- Melting Temperature: 28–32 °C, aligning with human comfort range.
- Encapsulation Material: Polymer or textile-based pouches were considered to prevent PCM leakage and facilitate easy integration.
- Outer Fabric: Lightweight, breathable polyester-cotton blends were selected to enhance airflow and wearer comfort.
- Other Parameters: High latent heat, non-toxicity, chemical stability, ease of recharge, reusability, and compatibility with clothing.

This phase ensured that the materials chosen could provide sustained cooling without compromising safety or comfort.

3.4: Questionnaire Survey

A pilot study was conducted using a structured questionnaire targeting 10 outdoor workers, including road workers, traffic police, MGNREGA workers, and delivery personnel. The survey focused on:

- Levels of physical exertion during work.
- Frequency of sweating and heat-related discomfort.
- Effectiveness and comfort of current uniforms.
- Perceived need and willingness to use a cooling vest.
- Preferences for vest weight, coverage area, and washability.

Responses were rated on a Likert scale (1–5) to facilitate quantitative analysis. Data from this study informed the ergonomic and functional design of the cooling vest and provided a baseline for evaluating user satisfaction.

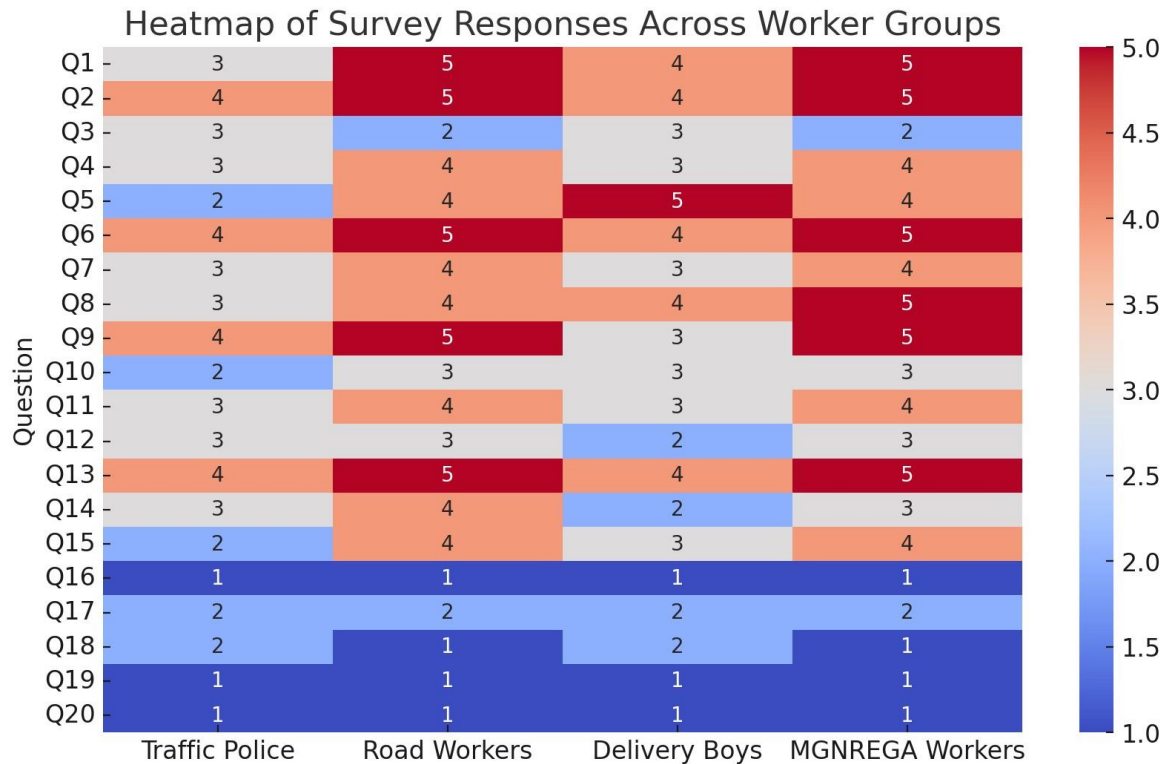


Fig 3.1 Survey result

3.5: Design and Modelling

The vest was modeled in SolidWorks using anthropometric data to ensure optimal fit and comfort.

Design Features:

- PCM Pocket Distribution: Strategically placed on chest, back, and sides for uniform cooling.
- Adjustable Straps and Panels: Allowing customization and improved mobility.
- Lightweight Design: Ensuring balanced mass distribution and minimal fatigue.

Thermal Simulation:

- Analysis of heat transfer during PCM phase change.
- Temperature variation across the vest surface.
- Duration of effective cooling before PCM fully melts.

This phase combined ergonomic design principles with thermal performance simulation, ensuring a balance between functionality and comfort.

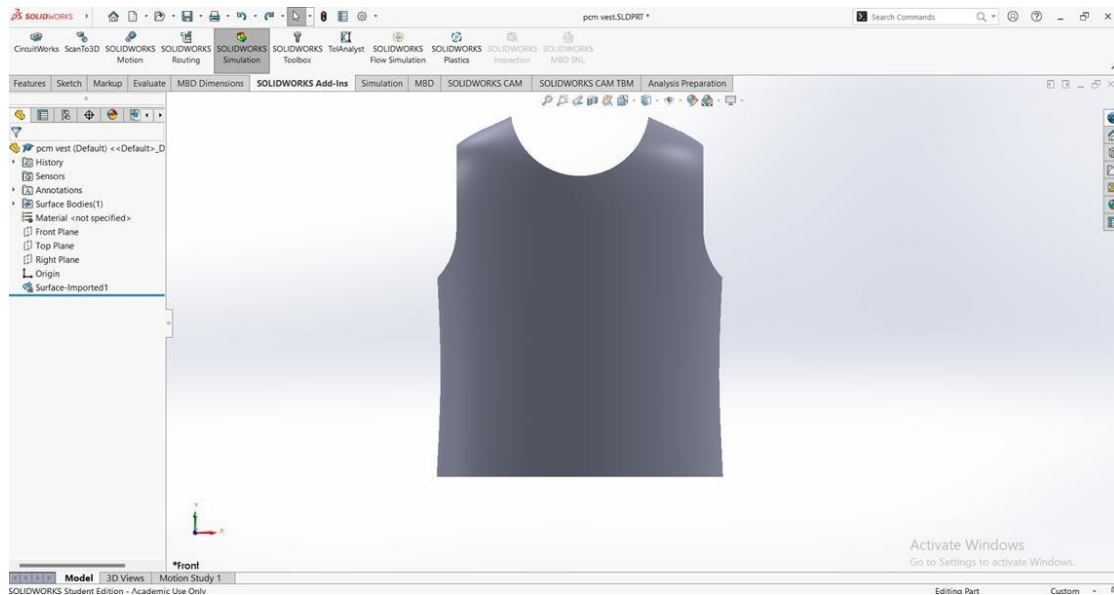


Fig 3.2 Design of vest using SolidWorks

3.6: Ergonomic Evaluation

Ergonomic considerations were integrated during both design and evaluation:

- Fit, flexibility, freedom of movement, and weight balance were assessed.
- Feedback from field users guided iterative adjustments to ensure comfort and usability.
- Posture analysis and user trials helped optimize the vest for different body types and movements.

This ensured the final design was practical for real-world applications, providing comfort without restricting movement.

3.7: Performance Evaluation (Simulation & Analysis)

Performance evaluation focused on both thermal and user-centered parameters:

- Cooling Effectiveness: Ability to maintain body temperature within 28–32°C.
- Duration of Cooling: Time until PCM completely melts.
- User Comfort: Derived from pilot survey ratings and feedback.
- Durability: Simulated repeated use and thermal cycling to estimate lifespan and longterm performance.

This phase validated the vest’s functional performance and informed any necessary design improvements.

3.8: Tools and Software Used

The following tools were employed throughout the project:

- SolidWorks: 3D modelling and ergonomic design.
- ANSYS Fluent: Thermal and structural simulations.
- CFD: Heat transfer analysis.
- EXCEL: Data analysis and visualization.
- SPSS: Statistical analysis of pilot survey results.

These tools enabled rigorous simulation, analysis, and validation of the vest design and performance.

- Reliable, reusable PCM performance over multiple cycles.
- Data supporting design validation and potential industrial-scale production.

The outcomes were expected to demonstrate both practical utility and commercial

CHAPTER 4

ERGONOMIC DESIGN AND MATERIAL SELECTION

4.1 Overview of Cooling Vest Design:

The design of the PCM-based cooling vest focuses on developing a portable and passive cooling system capable of reducing heat stress experienced by outdoor workers in hot environments. Workers such as road construction labourers, traffic police, delivery personnel, and rural employment workers are frequently exposed to high temperatures for prolonged periods. Continuous exposure to heat can cause thermal discomfort, dehydration, fatigue, and heat-related illnesses. Therefore, the development of a wearable cooling solution is necessary to maintain thermal comfort and improve work efficiency.

The cooling vest designed in this project utilizes **Phase Change Materials (PCM)** to absorb excess body heat. PCM materials store thermal energy during the **phase transition process from solid to liquid**, maintaining a relatively constant temperature during melting. This property makes PCM highly suitable for wearable cooling systems. The vest was designed as a **multi-layer structure** consisting of an inner comfort layer, a middle PCM cooling layer, and an outer insulation layer. Each layer performs a specific function that contributes to the overall thermal performance of the vest.

The design also considers ergonomic factors, ensuring that the vest is lightweight, flexible, and comfortable for long-term use. The PCM packs are distributed evenly across the vest to ensure uniform cooling across the torso region, which is one of the major heat exchange areas of the human body.

4.2 Material Selection for Cooling Vest:

The performance and effectiveness of a cooling vest largely depend on the materials used in its construction. Proper material selection is essential to ensure thermal comfort, durability, flexibility, and effective heat transfer between the human body and the cooling medium. In this project, the cooling vest consists of multiple layers, each serving a specific functional role such as comfort, cooling, and insulation. Therefore, materials were selected carefully based on their thermal properties, mechanical characteristics, and suitability for wearable applications.

The cooling vest developed in this study consists of three main layers: an inner comfort layer that comes into contact with the human body, a middle layer containing the phase change material (PCM) responsible for absorbing heat, and an outer protective layer that minimizes the influence of external environmental temperature. Each layer requires materials with different thermal and physical characteristics to ensure proper functioning of the cooling vest.

The inner layer is designed to provide comfort and efficient heat transfer from the human body to the PCM layer. Materials used for this layer should be breathable, lightweight, moisture-wicking, and soft to ensure user comfort during prolonged usage. Fabrics such as polyester-based moisture management textiles are commonly used for this purpose.

The middle layer contains the PCM packs which absorb heat from the body during the phase change process. This layer must be designed to securely hold PCM materials while allowing efficient thermal interaction with the inner layer.

The outer layer acts as a protective and insulating barrier against the external environment. It prevents excessive heat from entering the vest while also protecting the PCM packs from mechanical damage. Durable fabrics with moderate thermal insulation properties are typically used for the outer layer.

Proper material selection ensures that the vest can effectively transfer heat from the human body to the PCM layer while maintaining comfort and durability during use.

4.2.1 Selection Criteria:

Several factors were considered while selecting materials for the cooling vest to ensure effective thermal performance and user comfort. The most important selection criteria include thermal performance, comfort, durability, and safety.

Thermal conductivity is one of the key parameters that determines the rate at which heat is transferred from the body to the PCM layer. Materials with moderate thermal conductivity are preferred for efficient heat transfer without causing thermal discomfort.

Breathability is another important factor, especially for the inner layer of the vest. Since the vest is worn close to the human body, the fabric must allow air circulation and moisture evaporation to prevent excessive sweating and discomfort.

Weight and flexibility are also critical considerations. The vest should be lightweight and flexible so that it does not restrict body movement during work activities. Materials with high flexibility ensure that the vest conforms to the body shape and maintains good contact with the skin.

Durability and mechanical strength are important for the outer layer of the vest. The material should be able to withstand regular wear and tear, as well as environmental exposure.

Safety and non-toxicity of materials are also essential, especially for wearable cooling systems. The materials used should not cause skin irritation and must be safe for prolonged contact with the human body.

Cost and availability were also considered during material selection to ensure that the cooling vest design remains practical and economically feasible for large-scale use.

4.2.2 Thermal Properties of Materials:

The thermal properties of the materials used in the cooling vest play a crucial role in determining the overall cooling performance. These properties influence how effectively heat is transferred, stored, and dissipated within the vest system.

Thermal conductivity determines how easily heat can pass through a material. Materials with higher thermal conductivity allow faster heat transfer from the body to the PCM layer, improving cooling performance. However, extremely high conductivity may result in rapid heat loss once the PCM has melted, so a balanced value is desirable.

Specific heat capacity represents the amount of heat required to raise the temperature of a material. Materials with higher specific heat capacity can store more thermal energy, which helps in maintaining temperature stability within the vest.

Density of the material affects the thermal mass of the vest. Materials with higher density can store more thermal energy but may increase the overall weight of the vest. Therefore, an optimal balance between density and weight must be maintained.

Thermal insulation properties are important for the outer layer of the vest to prevent external heat from entering the cooling system. Insulating materials reduce the heat gain from the surrounding environment, thereby extending the cooling duration of the PCM.

In this project, polyester-based fabrics were selected for the inner and outer layers due to their durability, flexibility, and moderate thermal conductivity, while paraffin wax was selected as the PCM due to its excellent latent heat storage capability and suitable melting temperature range.

4.3 Properties of Phase Change Material:

Phase Change Materials (PCMs) are substances that absorb or release large amounts of heat during the process of changing their physical state, typically from solid to liquid or vice versa. This phase change occurs at a nearly constant temperature, allowing the PCM to maintain a stable thermal environment for a certain period.

In cooling vest applications, PCM absorbs heat from the human body as it melts, thereby reducing the body temperature and improving thermal comfort. Once the PCM has completely melted, the cooling effect gradually decreases as the material begins to store heat in the form of sensible heat.

PCMs are widely used in thermal management systems because of their ability to store large amounts of thermal energy in a compact volume. They provide passive cooling without the need for electrical power, making them suitable for wearable cooling applications.

PCMs are generally classified into three main types:

1. Organic PCMs
2. Inorganic PCMs
3. Eutectic PCMs

Organic PCMs such as paraffin wax are widely used in wearable cooling systems due to their chemical stability, non-corrosive nature, and reliable phase change behavior.

4.3.1 Paraffin Wax Characteristics

Paraffin wax is one of the most commonly used organic phase change materials for thermal energy storage applications. It is composed of a mixture of hydrocarbon compounds derived from petroleum. Paraffin wax is widely used in cooling systems because it possesses several favorable properties for thermal management.

One of the key advantages of paraffin wax is its stable and predictable melting behavior. It melts and solidifies repeatedly without significant degradation, making it suitable for repeated cooling cycles.

Paraffin wax is chemically stable and non-toxic, which makes it safe for use in wearable cooling systems. It also does not react with most common materials, reducing the risk of corrosion or material degradation.

Another advantage is its relatively high latent heat storage capacity. This allows paraffin wax to absorb a significant amount of heat during the phase change process without a large temperature increase.

Additionally, paraffin wax has low vapor pressure and minimal volume change during phase transition, which helps maintain the structural integrity of PCM containers. Due to these properties, paraffin wax is considered an ideal PCM for cooling vest applications.

4.3.2 Melting Temperature Range:

The melting temperature of the PCM is one of the most important parameters in cooling vest design. The PCM must have a melting temperature close to the human skin comfort temperature in order to effectively absorb heat from the body.

For wearable cooling applications, the ideal PCM melting temperature typically lies between 30°C and 34°C. This temperature range corresponds to approximately 303 K to 307 K, which is close to the human skin temperature.

When the PCM temperature reaches this melting range, it begins to absorb heat from the body and undergo a phase change from solid to liquid. During this process, the temperature remains nearly constant while the PCM absorbs a large amount of heat.

This constant temperature region is known as the phase change plateau and represents the effective cooling period of the vest.

In this project, paraffin wax with a melting temperature range of **30–34°C** was selected to ensure optimal cooling performance and thermal comfort for the user.

4.3.3 Latent Heat Storage Capacity:

Latent heat is the amount of heat energy absorbed or released by a substance during a phase change without a change in temperature. In PCM cooling systems, latent heat storage is responsible for the majority of the cooling effect.

Paraffin wax typically has a latent heat of fusion ranging between 200 kJ/kg and 220 kJ/kg. This means that a large amount of heat can be absorbed by the PCM while maintaining a nearly constant temperature.

When the cooling vest is worn, heat generated by the human body is transferred to the PCM layer. As the PCM begins to melt, it absorbs heat in the form of latent heat, thereby preventing the temperature of the vest from rising rapidly.

The high latent heat capacity of paraffin wax enables the cooling vest to provide effective thermal regulation for several hours, depending on the quantity of PCM used and the environmental conditions. This latent heat storage mechanism is the primary reason why PCM-based cooling systems are highly effective for wearable thermal management applications.

4.4 Multi-Layer Vest Structure:

The cooling vest consists of a three-layer configuration, where each layer plays a crucial role in thermal regulation and user comfort. The multilayer structure improves the overall cooling efficiency by controlling heat transfer from the human body to the surrounding environment.

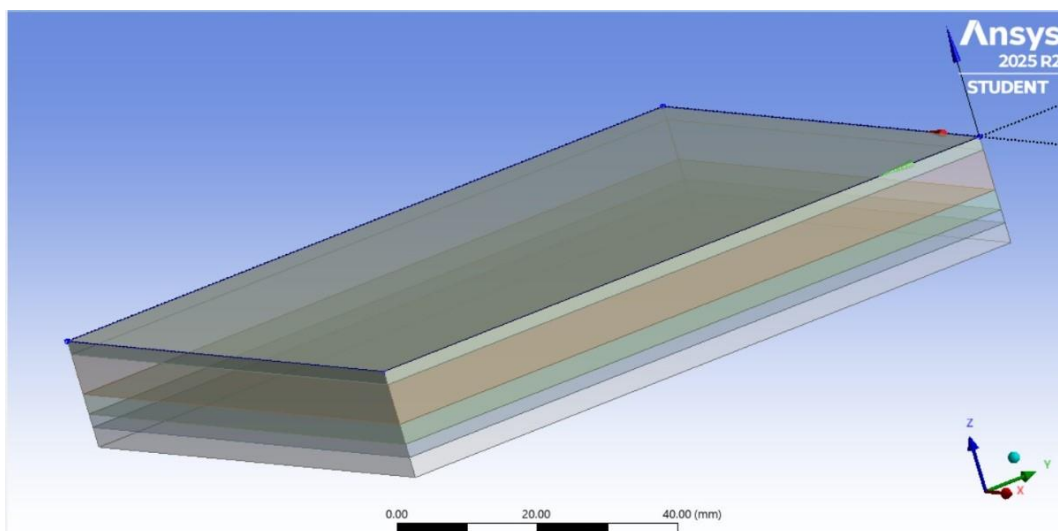


Fig 4.1: PCM vest structure

4.4.1 Inner Layer (Comfort and Moisture Management Layer)

The inner layer of the cooling vest is the layer that directly contacts the human skin. The primary purpose of this layer is to provide comfort while facilitating efficient heat transfer from the body to the PCM layer. In this project, Coolmax polyester fabric is selected as the inner layer material due to its excellent moisture-wicking and breathability properties.

Coolmax polyester fabric is engineered with specially designed fibers that allow sweat to be quickly transported away from the skin surface. This helps maintain dryness and prevents discomfort caused by sweat accumulation. Additionally, the fabric is lightweight, flexible, and breathable, making it suitable for prolonged usage in outdoor conditions.

Another important function of the inner layer is to act as a heat transfer medium between the human body and the PCM cooling layer. Since the fabric has moderate thermal conductivity, it allows body heat to pass through efficiently without causing excessive heat retention near the skin. This ensures that the cooling effect of the PCM is effectively transferred to the body.

4.4.2 Middle Layer (PCM Cooling Layer):

The middle layer of the vest contains the Phase Change Material (PCM) which is responsible for the primary cooling effect. In this project, paraffin wax is used as the PCM due to its favorable thermal properties and safety characteristics.

Paraffin wax is widely used in thermal energy storage applications because it possesses a high latent heat capacity, meaning it can absorb a significant amount of heat during the phase change process. The PCM used in the vest has a melting temperature range between 30°C and 34°C, which is suitable for maintaining human thermal comfort.

During operation, body heat from the user is transferred through the inner fabric layer to the PCM packs. As the temperature of the PCM reaches its melting point, the material begins to change from solid to liquid while absorbing a large amount of heat energy. During this phase transition, the temperature of the PCM remains nearly constant, which allows the vest to maintain a stable cooling temperature for an extended period.

The PCM is contained within sealed pouches or compartments distributed evenly across the vest. This arrangement ensures uniform heat absorption and prevents localized overheating. The PCM layer is the core functional component of the cooling vest.

4.4.3 Outer Layer (Thermal Insulation and Protection Layer):

The outermost layer of the cooling vest is designed to protect the PCM layer and reduce the impact of external environmental heat. In this project, Polyester Oxford fabric is selected as the outer layer material due to its durability, water resistance, and insulating properties.

Polyester Oxford fabric is commonly used in outdoor garments and protective clothing because it provides excellent mechanical strength and resistance to wear and tear. The fabric also has moderate thermal insulation properties that help reduce heat transfer from the surrounding environment to the PCM layer.

By minimizing external heat penetration, the outer layer ensures that the PCM primarily absorbs heat from the human body rather than the surrounding environment. This improves the efficiency of the cooling system and extends the cooling duration of the vest.

4.5 Ergonomic Design Considerations:

Ergonomic design is a critical aspect of wearable cooling systems. A cooling vest must provide thermal benefits without compromising comfort, mobility, or worker productivity. Therefore, several ergonomic factors were considered during the design of the cooling vest.

The vest dimensions were determined based on human anthropometric measurements of the upper body, including chest circumference, shoulder width, and torso length. These measurements ensure that the vest fits properly across different body sizes and provides uniform cooling coverage.

Weight distribution is another important ergonomic factor. PCM materials add additional weight to the vest, and uneven weight distribution can cause discomfort or fatigue. To address this issue, the PCM packs were arranged symmetrically across the front and back of the vest to maintain balance.

The vest is also designed with flexible fabric materials and adjustable straps, allowing users to move freely during physical activities such as walking, lifting, and bending. Proper ergonomic design ensures that the cooling vest enhances worker comfort without interfering with daily work tasks.

4.6 Summary

This chapter presented the ergonomic design considerations and material selection for the PCM-based cooling vest. The materials used in the vest were selected based on criteria such as thermal conductivity, comfort, breathability, durability, and safety.

A multilayer cooling vest structure consisting of an inner comfort layer, a PCM cooling layer, and an outer protective layer was proposed. The thermal properties of the selected materials play an important role in determining the heat transfer performance and cooling efficiency of the vest.

Paraffin wax was chosen as the phase change material due to its suitable melting temperature range, high latent heat storage capacity, chemical stability, and non-toxic nature. The melting temperature range of 30–34°C allows the PCM to effectively absorb heat from the human body while maintaining thermal comfort.

The properties of paraffin wax make it an ideal material for wearable cooling systems, enabling efficient thermal energy storage and prolonged cooling duration. The next chapter focuses on the thermal modeling and simulation of the cooling vest using numerical analysis tools.

CHAPTER 5

THERMAL ANALYSIS OF MULTILAYER COOLING VEST USING ANSYS FLUENT

5.1 Introduction

Thermal analysis is an essential step in evaluating the performance of a phase change material (PCM) based cooling vest. In this study, numerical thermal analysis was performed using **ANSYS Fluent**, which is a computational fluid dynamics (CFD) software widely used for solving heat transfer, fluid flow, and phase change problems.

Workers such as construction labourers, traffic police personnel, delivery workers, and outdoor maintenance staff are often exposed to extreme environmental temperatures. Prolonged exposure to high temperatures can lead to thermal discomfort, dehydration, fatigue, and heat-related illnesses. Cooling garments incorporating phase change materials provide a passive cooling mechanism that helps regulate body temperature and improve worker comfort.

Phase change materials have the ability to absorb large amounts of heat during their phase transition from solid to liquid. During this process, the PCM maintains a nearly constant temperature while storing thermal energy in the form of latent heat. This property makes PCM highly suitable for wearable cooling applications.

The purpose of conducting thermal simulation in this project is to evaluate the cooling performance of the PCM layer integrated within the cooling vest. The simulation allows the prediction of temperature distribution across the vest layers and helps understand the heat transfer mechanism between the human body and the PCM material.

The thermal analysis helps to:

- Understand the heat transfer behaviour within the multilayer vest structure
- Determine the temperature distribution across different layers of the vest
- Study the melting behaviour of the PCM material
- Estimate the cooling duration provided by the PCM
- Validate the effectiveness of the proposed cooling vest design

Instead of directly conducting experimental trials, simulation provides a cost-effective and efficient method for analysing thermal performance under controlled conditions. The results obtained from ANSYS Fluent provide valuable insight into the PCM melting process and the cooling effect generated by the vest.

5.2 Geometry Development of Multilayer Cooling Vest Model

In real conditions, the cooling vest consists of multiple fabric layers and PCM pockets distributed across the torso region. However, modelling the complete vest geometry would require extensive computational resources and complex modelling procedures. Therefore, for the purpose of thermal analysis, the vest structure was simplified into a **multilayer block model**.

The simplified multilayer block represents the thermal behaviour of the cooling vest while reducing computational complexity. Each layer of the vest is represented as a separate region in the model.

The following layers were included in the simulation model:

Layer	Description
Skin layer	Represents the human body heat source
Inner fabric layer	Fabric layer in contact with skin
Air gap	Insulating region between layers
PCM layer	Heat absorbing layer
Outer fabric layer	Protective outer layer

Table 5.1:Description of Multiple Layers

The multilayer configuration replicates the actual heat transfer path occurring in the cooling vest.

Heat generated by the human body travels through the inner fabric layer and the air gap before reaching the PCM layer. The PCM absorbs this heat energy during the phase change process. The outer layer protects the PCM pockets and interacts with the surrounding environment.

The geometry used in the simulation was a **rectangular block of dimensions 100 mm × 100 mm**, representing a small section of the cooling vest.

The thickness of each layer was chosen based on typical garment construction and PCM pouch dimensions.

Layer	Thickness
Skin layer	3 mm
Inner fabric layer	2 mm
Air gap	3 mm
PCM layer	10 mm
Outer fabric layer	2 mm

Table 5.2: Thickness of multiple Layers

This simplified multilayer structure allows efficient simulation while accurately capturing the key heat transfer processes occurring in the cooling vest.

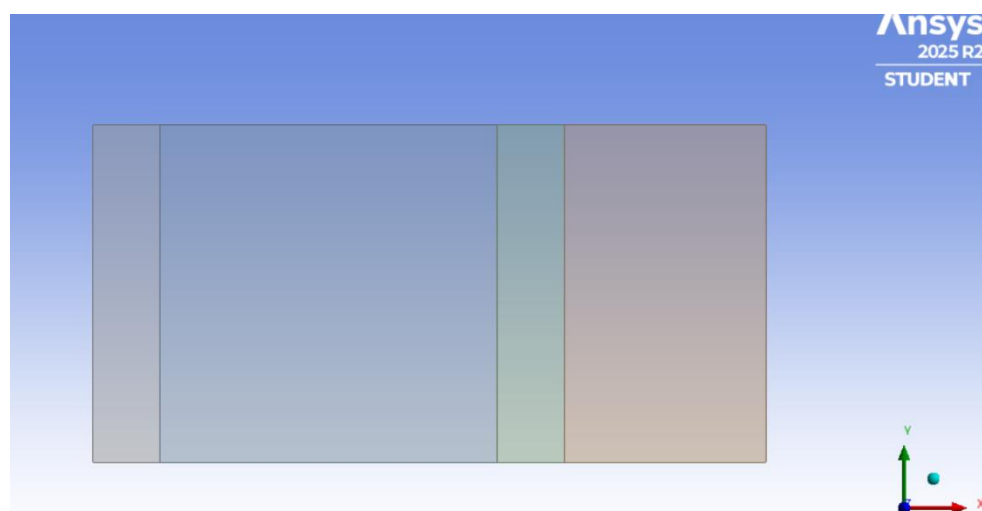


Fig 5.1: 2D Model

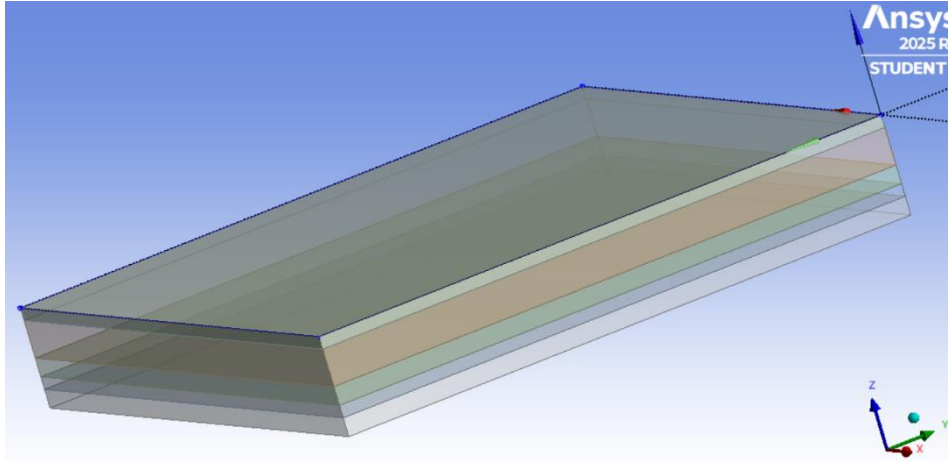


Fig 5.2: 3D Model

5.3 Mesh Generation

Meshing is a critical step in computational simulations because it converts the continuous geometry into small discrete elements where numerical equations can be solved.

In this study, the geometry was discretized using the **ANSYS Meshing module** before importing the mesh into ANSYS Fluent.

A structured mesh was used to improve the accuracy and convergence stability of the simulation. The mesh was generated using quadrilateral elements in the two-dimensional model.

Special attention was given to the interfaces between different material layers because temperature gradients are expected to be higher at these boundaries.

To capture the heat transfer behaviour more accurately, mesh refinement was performed near the PCM layer. Finer mesh elements help in accurately tracking the melting process of the PCM material.

The key mesh parameters used in the simulation are:

Average element size: 1–2 mm

Total number of elements: approximately 8000 – 12000

Element type: quadrilateral elements

Inflation layers were applied near material interfaces to capture boundary layer heat transfer effects.

Mesh independence analysis was also considered to ensure that further mesh refinement does not significantly affect the simulation results. This ensures that the results obtained are reliable and not dependent on mesh size.

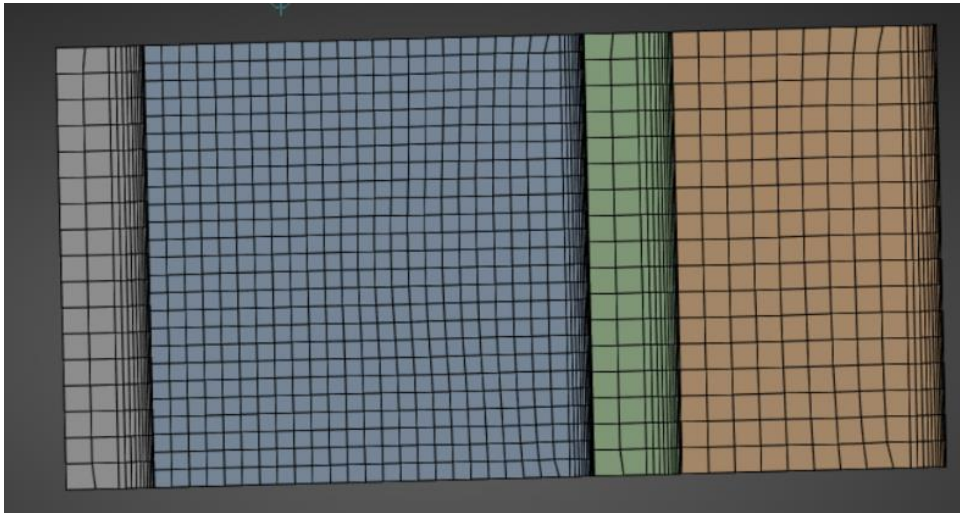


Fig 5.3: 2D Meshing

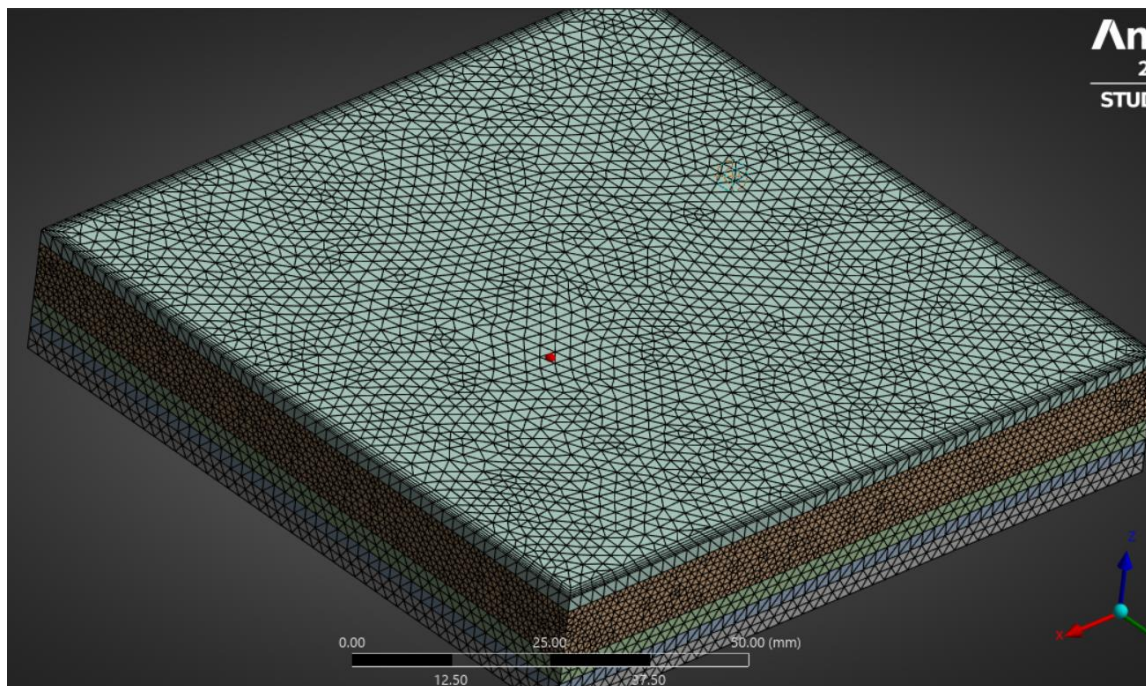


Fig 5.4: 3D Meshing

5.4 Material Properties

Material properties play an important role in determining the thermal performance of the cooling vest. Each layer in the multilayer block was assigned appropriate thermophysical properties.

The materials used in the simulation include human skin, Coolmax polyester fabric, air, PCM material, and polyester Oxford fabric.

Skin Layer

The skin layer represents the heat source from the human body. The average surface temperature of the human body is approximately 310 K (37°C).

Inner Fabric Layer

The inner layer of the vest is made of Coolmax polyester fabric, which is known for its moisture-wicking and breathable properties. This fabric improves comfort and allows sweat evaporation.

Air Gap

A small air gap exists between the fabric layers, which acts as an insulating region. Air has low thermal conductivity, which helps reduce heat transfer to the environment.

PCM Layer

The PCM layer is the most important component of the cooling vest. In this project, paraffin wax was selected as the PCM due to its suitable melting temperature range and high latent heat capacity.

Outer Fabric Layer

The outer layer is made of polyester Oxford fabric, which provides durability and protection against environmental exposure.

The thermophysical properties used in the simulation are shown below:

Material	Density (kg/m ³)	Thermal Conductivity (W/mK)	Specific Heat (J/kgK)
Skin	1100	0.37	3500
Coolmax polyester	1380	0.04	1200
Air	1.225	0.024	1006
Paraffin wax	900	0.21	2000
Polyester Oxford	1300	0.05	1300

Table 5.3: Thermophysical Properties of Materials

PCM phase change properties used:

Melting temperature range = **303 K – 305 K**

Latent heat of fusion = **200 kJ/kg**

These properties enable the PCM to absorb significant amounts of heat while maintaining a stable temperature during the melting process.

5.5 Boundary Conditions

Boundary conditions define the physical environment in which the cooling vest operates.

The following boundary conditions were applied in the simulation:

Human skin temperature = **310 K**

Ambient environmental temperature = **308 K**

Initial PCM temperature = **298 K**

The outer surface of the vest was exposed to the surrounding air. Heat exchange between the outer layer and the environment occurs through natural convection.

The convection heat transfer coefficient was assumed to be approximately:

$h = 10 \text{ W/m}^2\text{K}$

These conditions represent typical hot outdoor working environments.

5.6 Simulation Setup in ANSYS Fluent

The thermal analysis was performed using a **transient solver** because the PCM melting process is time dependent.

The following solver settings were used in ANSYS Fluent:

Solver type: Pressure-based solver

Time formulation: Transient

Energy equation: Enabled

Solidification and melting model: Activated

The PCM melting process was simulated using the enthalpy-porosity approach available in Fluent.

The time step size was selected carefully to ensure numerical stability and accurate prediction of the melting process.

Time step size = **1 seconds**

Iterations per time step = **20**

Total simulation time = **18000 seconds**

This allows the simulation to capture the entire PCM melting process and evaluate the cooling duration provided by the vest.

5.7 Validation Methodology of the Simulation Model

Validation is an important step in computational studies to ensure that the simulation results represent realistic physical behaviour. In this project, the numerical simulation was validated through a simplified modelling approach and comparison with expected thermal behaviour of PCM materials.

The validation process involved analysing the temperature response and melting characteristics of the PCM layer under controlled conditions.

A simplified **two-dimensional multilayer block model** was first developed to represent a section of the cooling vest. The simplified geometry allowed efficient simulation while maintaining the essential thermal characteristics of the system.

The validation methodology consisted of the following steps:

Step 1: Simplified Geometry Creation

A rectangular multilayer block with dimensions **100 mm × 100 mm** was created. The block consisted of the following layers:

- Skin layer
- Inner fabric layer
- Air gap
- PCM layer
- Outer fabric layer

This simplified structure represents the thermal pathway between the human body and the surrounding environment.

Step 2: Assignment of Material Properties

Thermophysical properties for each layer were obtained from published literature and engineering databases. These properties include:

- Density
- Thermal conductivity
- Specific heat capacity
- Latent heat of fusion (for PCM)

Accurate material properties are essential for reliable thermal simulations

Step 3: Application of Boundary Conditions

Boundary conditions were applied to represent realistic working conditions.

Human skin temperature = **310 K**

Ambient temperature = **308 K**

Initial PCM temperature = **298 K**

These conditions simulate a hot outdoor environment where workers are exposed to elevated temperatures.

Step 4: Transient Thermal Simulation

The simulation was conducted using a transient solver in ANSYS Fluent to capture the time-dependent behaviour of the PCM melting process.

The simulation parameters included:

Time step size = **10 seconds**

Iterations per time step = **20**

Total simulation duration = **18000 seconds**

The transient simulation allowed observation of temperature evolution and PCM melting behaviour over time.

Step 5: Analysis of Temperature Profiles

Temperature profiles were extracted from different locations within the multilayer block. These temperature variations were plotted against time to analyse the cooling behaviour of the PCM layer.

The simulation results showed three distinct thermal regions:

1. Initial heating stage
2. Phase change temperature plateau
3. Post-melting temperature rise

During the phase change stage, the PCM temperature remained nearly constant within the melting range of **303 K to 305 K**, which confirms the expected behaviour of paraffin-based PCM materials.

Step 6: Verification of PCM Cooling Duration

The simulation results indicated that the PCM maintained a stable temperature range for approximately **4 to 5 hours**, which corresponds to the duration of the melting process.

This result is consistent with reported experimental studies on PCM-based cooling garments.

Step 7: Comparison with PCM Thermal Behaviour

The simulation results were compared with typical thermal characteristics of paraffin wax PCM reported in literature.

The following behaviours matched the expected PCM characteristics:

- Temperature plateau during phase change
- Gradual increase in liquid fraction
- Stable cooling duration during melting

These results confirm that the simulation model accurately captures the phase change behaviour of the PCM layer.

Summary of Validation

The validation process confirmed that the ANSYS Fluent simulation model successfully represents the thermal behaviour of PCM in the cooling vest.

The results demonstrate that:

- The PCM absorbs heat effectively during phase change
- The temperature remains nearly constant during melting
- The cooling vest can provide prolonged thermal comfort

Therefore, the developed simulation model can be used to evaluate the performance of the PCM cooling vest design.

5.8 Results and Thermal Performance Analysis of PCM Cooling Vest

5.8.1 Introduction

This section presents the results obtained from the transient thermal analysis of the multilayer cooling vest using CFD simulations performed in Ansys Fluent. The objective is to evaluate

the thermal performance of the phase change material (PCM) in maintaining skin temperature within a comfortable range and to analyze its heat absorption characteristics over time.

5.8.2 Temperature Variation with Time

The variation of volume-averaged temperature with respect to flow time is shown in **Figure 5.5**. The graph indicates that the temperature remains nearly constant within the range of 303 K to 305 K for a significant duration, corresponding to the phase change region of the PCM.

This plateau region extends up to approximately 14,400 seconds, after which a sharp rise in temperature is observed. This increase indicates the completion of the PCM melting process and the reduction in its heat absorption capability.

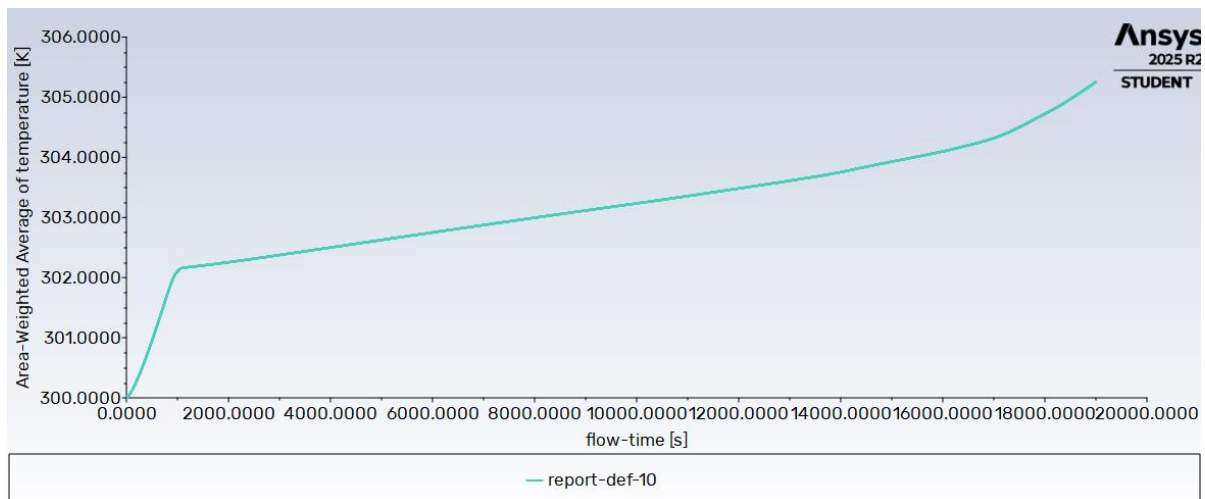


Figure 5.5

(Temperature vs Flow Time graph)

Inference:

The stable temperature plateau confirms that the PCM effectively regulates temperature by absorbing latent heat during phase transition.

5.8.3 Liquid Fraction Analysis

The variation of liquid fraction with flow time is illustrated in **Figure 5.6**. The liquid fraction increases gradually from 0 to 1, indicating the progressive melting of the PCM.

At around 18,000 seconds, the liquid fraction reaches unity, signifying complete melting of the PCM. Beyond this point, no further latent heat absorption occurs, resulting in a temperature rise.

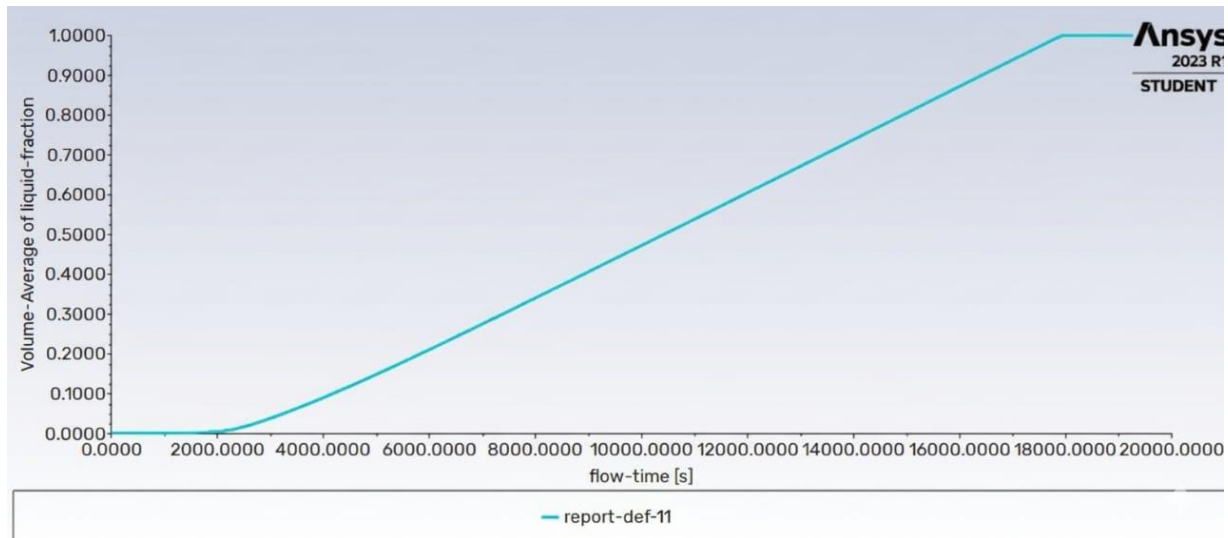


Figure 5.6
(Liquid fraction vs Flow Time graph)

Inference:

The liquid fraction plot validates the duration of effective cooling and confirms that the thermal performance is governed by the phase change process.

5.8.4 Temperature Contour Analysis

The temperature distribution across the multilayer cooling vest at different time intervals is shown in **Figures 5.7, 5.8, and 5.9**.

- **Figure 5.7** represents the initial stage, where the temperature is uniformly low across the PCM layer.
- **Figure 5.8** shows the intermediate stage, where heat penetration into the PCM layer begins and temperature gradients develop.

- **Figure 5.9** represents the final stage, where most of the PCM has melted, and higher temperatures are observed.

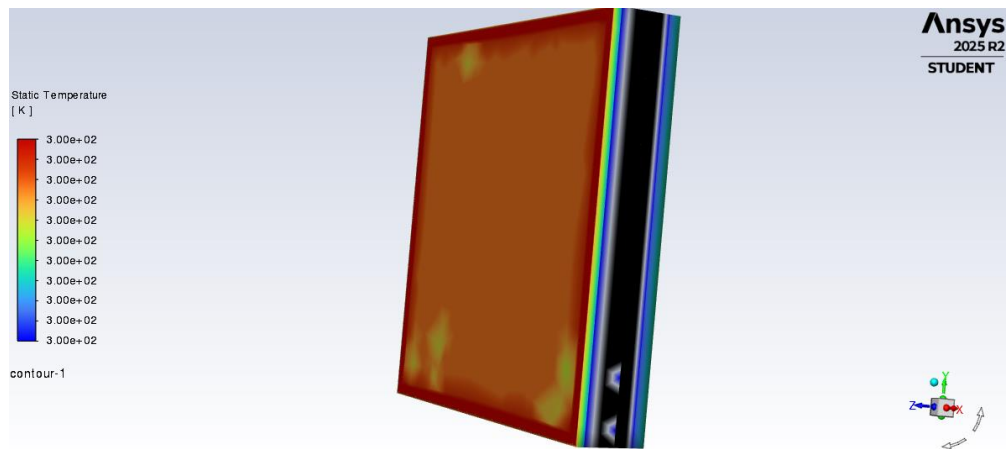


Figure 5.7 (*Initial temperature contour image*)

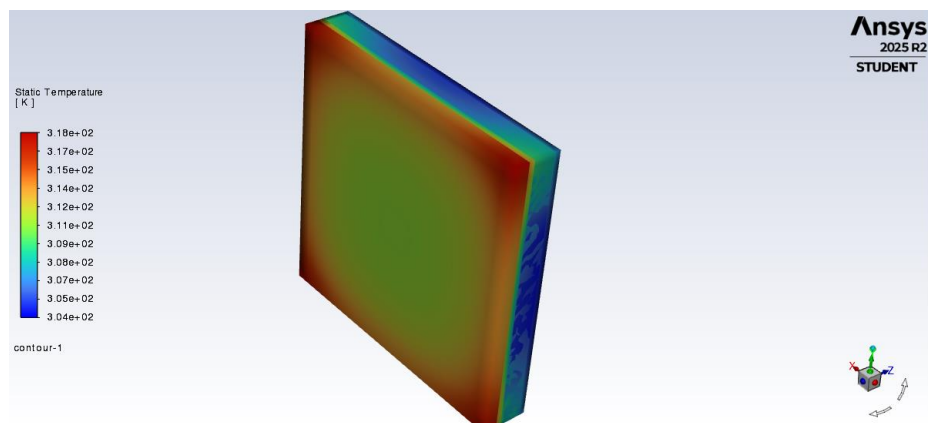


Figure 5.4

(*Intermediate temperature contour image*)

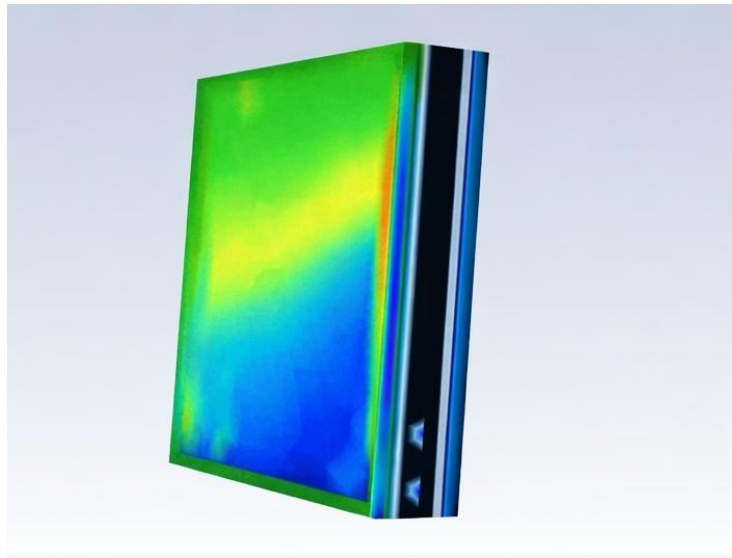


Figure 5.8
(Final temperature contour image)

Interpretation of the Two Temperature Contours

Fig 5.7 – Initial State (Before PCM absorbs heat)

Characteristics visible:

Outer layer → High temperature (red ~36–37°C)

Skin layer → Also warm

PCM layer → Still solid / not fully activated

Heat from body and environment has not yet been absorbed by PCM

Meaning:

This represents $t = 0$ s or early simulation time

PCM has not melted yet

Heat transfer has not started significantly

So this can be labeled as:

Initial Thermal State of Cooling Vest

Fig 5.8 – Final State (After PCM cooling effect)

Characteristics visible:

Skin side → Blue / light blue (~32–34°C)

Middle layer → Green (~34–35°C)

Temperature gradient across layers

PCM absorbing heat → latent heat effect

Meaning:

PCM is melting and absorbing body heat

Heat transfer from skin → PCM → outer environment

Skin temperature reduced

This represents:

Cooling performance after several hours

Example label:

Temperature Distribution after PCM Latent Heat Absorption

Layer	Temperature
Skin	36–37 °C
Inner Fabric	36 °C
Air Gap	35 °C
PCM	28–30 °C
Outer Layer	34–35 °C

Table 5.4 Initial Condition

Layer	Temperature
Skin	32–33 °C
Inner Fabric	33–34 °C
Air Gap	34 °C
PCM	29–30 °C
Outer Layer	35 °C

Table 5.5 Final Condition

Result:

Skin temperature reduction = 3–4 °C

Inference:

The contour plots demonstrate effective heat transfer from the skin surface to the PCM layer and illustrate the progression of the melting front over time.

5.8.5 Performance Evaluation

The key performance parameters obtained from the simulation are summarized below:

- **Skin Temperature:** 28 °C – 30 °C
- **Cooling Duration:** $\approx$ 5 hours
- **PCM Temperature Plateau:** 303 K – 305 K
- **Total Heat Absorbed:** 270 kJ

The PCM maintains the skin temperature within the comfort range throughout the cooling duration. The plateau region confirms thermal stability, while the high heat absorption capacity ensures prolonged cooling.

The average heat removal rate is approximately **15 W**, indicating effective passive cooling performance suitable for wearable applications.

5.8.6 Discussion

The results clearly indicate that the PCM-based cooling vest provides sustained thermal regulation. The temperature plateau observed during the phase change process ensures that the skin temperature does not rise rapidly, thereby enhancing user comfort.

The cooling duration of approximately 5 hours makes the system suitable for practical applications such as outdoor work, military operations, and industrial environments. However, once the PCM is fully melted, the cooling effect diminishes, leading to a gradual increase in temperature.

The contour and liquid fraction results collectively validate the effectiveness of the PCM and the accuracy of the simulation model developed in Ansys Fluent.

5.8.7 Summary

This section demonstrated that the multilayer PCM cooling vest is capable of maintaining thermal comfort through efficient latent heat absorption. The simulation results confirm stable temperature regulation, adequate cooling duration, and high heat storage capacity, validating the design and its performance.

CHAPTER 6

DESIGN OPTIMIZATION, PERFORMANCE EVALUATION, AND MARKET COMPARISON OF PCM COOLING VEST

6.1 Performance of Existing PCM Cooling Vests

Phase Change Material (PCM) cooling vests are widely used for reducing heat stress among outdoor workers, athletes, and personnel working in high-temperature environments. These vests utilize PCM packs that absorb body heat during the phase change process, thereby maintaining the body temperature within a comfortable range.

Commercial PCM cooling vests typically consist of multiple PCM packs placed inside pockets distributed across the vest. The PCM absorbs heat from the body and undergoes a phase change from solid to liquid, maintaining a nearly constant temperature during the melting process.

The typical performance characteristics of commercially available PCM cooling vests are summarized in Table 6.1.

Parameter	Typical Value
Total vest weight	1.5 – 2 kg
PCM quantity	0.9 – 2 kg
Cooling duration	1 – 4 hours
PCM inserts	4 – 8 packs

Table 6.1 Performance of Commercial PCM Cooling Vests

Some advanced PCM cooling vests are capable of providing up to four hours of cooling while weighing approximately 1.5 kg depending on the quantity of PCM used and the vest

design. Lightweight models typically contain about 900 g of PCM and weigh around 1.2–1.5 kg, but these designs usually provide shorter cooling durations of only 1–2 hours.

6.2 Limitations of Existing PCM Cooling Vests

Despite their advantages, currently available PCM cooling vests exhibit several limitations in terms of cooling efficiency, ergonomics, and cost.

1. Uneven Cooling Distribution

Most commercial cooling vests use 4–8 large PCM packs placed in limited locations. As a result, the cooling effect is concentrated only in certain areas of the torso, leaving other areas insufficiently cooled.

2. Bulkiness of PCM Packs

Large PCM packs increase the thickness of the vest, making the vest bulky and restricting body movement. This reduces comfort during long working hours.

3. Poor Ergonomic Fit

Many commercial designs do not maintain proper contact between the PCM pack and the body surface. Air gaps between the PCM packs and the body reduce the effectiveness of heat transfer.

4. High Cost

Commercial PCM cooling vests are relatively expensive and are often priced between **₹12,000 and ₹25,000**. This makes them inaccessible to many outdoor workers, especially in developing countries.

6.3 Proposed Ergonomic PCM Cooling Vest Design

To overcome the limitations of existing PCM cooling vests, the present study proposes an **ergonomically optimized PCM cooling vest with distributed PCM pockets**.

The design focuses on improving:

- Cooling uniformity
- Heat transfer efficiency
- Vest flexibility

- Cost effectiveness

The major improvements introduced in the proposed design include:

- Distributed PCM packs across the torso
- Reduced PCM pack thickness
- Improved thermal contact with the body
- Lower manufacturing cost

6.4 PCM Pocket Distribution

In the proposed design, PCM packs are distributed across the chest and back regions to achieve uniform cooling of the torso.

Location	Number of PCM Pockets
Chest	4
Upper Back	3
Lower Back	3
Total	10 pockets

Table 6.2 Distribution of PCM Pockets

This distributed configuration improves the cooling coverage across the torso and ensures better thermal comfort.

6.5 Optimization of PCM Pocket Dimensions

To improve flexibility and body contact, thin ergonomic PCM packs were selected instead of large bulky PCM blocks.

Parameter	Value
Length	150 mm
Width	100 mm
Thickness	10 mm

Table 6.3 Optimized PCM Pack Dimensions

Thin PCM packs offer several advantages:

- Increased flexibility of the vest
- Better body contact
- Improved heat transfer efficiency

6.6 Volume of PCM per Pocket

The volume of PCM required for each pocket is calculated using:

$$V = L \times W \times T$$

Where:

- L = length of PCM pack
- W = width of PCM pack
- T = thickness of PCM pack

$$V = 0.15 \times 0.10 \times 0.01$$

$$V = 0.00015 \text{ m}^3$$

Thus, the volume of PCM required per pocket is:

$$V = 0.00015 \text{ m}^3$$

6.7 PCM Mass per Pocket

The mass of PCM is calculated using the density relation:

$$m = \rho V$$

Where:

$$\rho = 900 \text{ kg/m}^3 \text{ (density of paraffin wax)}$$

$$m = 900 \times 0.00015$$

$$m = 0.135 \text{ kg}$$

Therefore, each pocket contains approximately:

110 g of PCM

6.8 Total PCM Required in the Vest

Since the vest contains 10 PCM pockets, the total PCM mass is calculated as:

$$\text{Total PCM} = 10 \times 0.135$$

$$\text{Total PCM} = 1.35 \text{ kg}$$

Thus, the vest contains approximately **1.35 kg of PCM**.

6.9 Cooling Capacity of PCM

The cooling capacity depends on the latent heat of fusion of the PCM.

For paraffin wax:

$$L = 200 \text{ kJ/kg}$$

The total heat absorption capacity is:

$$Q = mL$$

$$Q = 1.35 \times 200$$

$$Q = 270 \text{ kJ}$$

Thus, the PCM in the vest can absorb approximately **216 kJ of heat energy** during the phase change process.

6.10 Estimated Cooling Duration

The average heat load from the human torso during moderate activity is approximately **15 W**.

Cooling duration is calculated using:

$$t = \frac{Q}{P}$$

Where:

$$Q = 270000 \text{ J}$$

$$P = 15 \text{ W}$$

$$t = \frac{270000}{15}$$

$$t = 18000 \text{ seconds}$$

$$t \approx 5 \text{ hours}$$

Thus, the proposed cooling vest can provide approximately **5 hours of cooling**.

6.11 Total Vest Weight

Component	Weight
PCM	1.35 kg
Fabric layers	0.4 kg
Total Weight	1.75 kg

Table 6.4 Estimated Vest Weight

The total vest weight is approximately **1.75 kg**, which is comparable to or lighter than many commercial PCM cooling vests.

6.12 Cost Analysis

The estimated manufacturing cost of the proposed vest is shown in Table 5.5.

Component	Cost
Paraffin PCM (1.1 kg)	₹430
Fabric layers	₹400
Pocket stitching	₹250
Encapsulation pouch	₹200
Total Cost	₹1280

Table 6.5 Estimated Manufacturing Cost

Commercial PCM cooling vests are priced between **₹12,000 and ₹25,000**.

Therefore, the proposed vest reduces cost by nearly **90%**, making it suitable for large-scale use among outdoor workers.

6.13 Ergonomic Advantages of the Proposed Design

The proposed cooling vest provides several ergonomic benefits:

1. Better Weight Distribution

The use of **10 smaller PCM packs** distributes the weight evenly across the torso, reducing localized pressure.

2. Improved Flexibility

Thin PCM packs of **8 mm thickness** allow better movement compared to bulky PCM blocks.

3. Enhanced Heat Transfer

Increased surface contact between the PCM packs and the body improves the efficiency of heat transfer.

4. Reduced Vest Weight

The optimized PCM configuration reduces the overall vest weight while maintaining effective cooling duration.

6.14 Comparison with Existing PCM Cooling Vests

Feature	Market Vest	Proposed Design
PCM packs	4–8	10 distributed packs
PCM mass	0.9–2 kg	1.35 kg optimized
Vest weight	1.5–2 kg	1.75 kg
Cooling duration	1–4 hours	≈ 5 hours
Cost	₹12,000–₹25,000	₹1280
Ergonomics	Moderate	Improved flexibility

Table 6.6 Comparison of Proposed Vest with Commercial Cooling Vests

6.15 Ergonomic Design and PCM Pocket Distribution of Cooling Vest

The ergonomic design of the Phase Change Material (PCM)-based cooling vest was developed with the primary objective of enhancing thermal comfort while ensuring maximum mobility, flexibility, and user convenience for outdoor workers. The design was created using SolidWorks by incorporating standard anthropometric data, allowing the vest to fit a wide range of body types without restricting movement.

Unlike conventional cooling vests, which often suffer from uneven cooling and poor weight distribution, the proposed design emphasizes a balanced, lightweight, and functionally optimized structure. The vest follows a sleeveless configuration with widened arm openings and a contoured neckline, ensuring that the user can perform dynamic activities such as walking, bending, lifting, and reaching without discomfort.

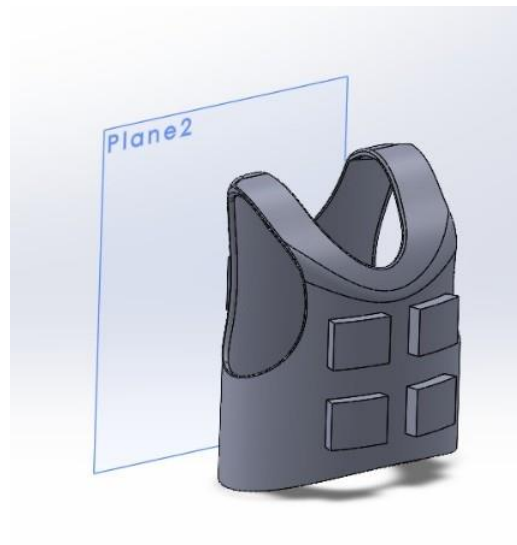
6.15.1 SolidWorks-Based Ergonomic Design

The 3D model of the cooling vest was developed using SolidWorks to visualize the placement of PCM pockets and evaluate the overall structural design. The modelling process involved designing the vest geometry, defining the multi-layer structure, and integrating PCM pockets at ergonomically suitable locations.

The vest was designed to closely follow the natural curvature of the human torso to ensure proper contact between the vest and the body. This is critical for effective heat transfer from the body to the PCM layer. The side regions of the vest were kept flexible and free from rigid components to allow natural body movements without restriction.

Additionally, the thickness of the vest was optimized to ensure that the PCM pockets do not create excessive bulk, thereby maintaining user comfort during prolonged usage.

Figure 6.1: SolidWorks Front View of PCM Cooling Vest with 4 PCM Pockets



6.15.2 Front Side PCM Pocket Configuration (4 Pockets)

The front portion of the vest consists of four PCM pockets arranged symmetrically across the chest region. This placement is strategically selected based on the physiological importance of the chest area in heat exchange.

The chest region contains vital organs such as the lungs and heart, which contribute significantly to metabolic heat generation. By placing PCM pockets in this region, the vest effectively absorbs heat from the core body area, resulting in faster cooling and improved thermal comfort.

The pockets are designed in a rectangular shape and positioned with adequate spacing between them. This spacing is crucial to:

- Maintain flexibility of the vest
- Prevent stiffness during forward bending
- Ensure even distribution of cooling effect
- Avoid excessive localized cooling

The symmetrical arrangement ensures that the cooling effect is uniform across both sides of the body, thereby improving comfort and preventing imbalance.

6.15.3 Back Side PCM Pocket Configuration (6 Pockets)

The back portion of the vest incorporates six PCM pockets, distributed across the upper and lower back regions. The decision to place a higher number of pockets at the back is based on both thermal and ergonomic considerations.

The human back has a larger surface area compared to the chest and experiences significant heat accumulation during physical work. Activities such as lifting, bending, and prolonged standing increase heat generation in the back region. Therefore, providing additional PCM pockets in this area enhances the cooling performance of the vest.

The six pockets are arranged in a structured layout to:

- Cover maximum surface area of the back
- Ensure uniform cooling across upper and lower regions
- Increase total cooling capacity
- Extend cooling duration

This distribution also helps in reducing sweat accumulation and discomfort in the back region, which is a common issue among outdoor workers.



Figure 6.2: SolidWorks Back View of PCM Cooling Vest with 6 PCM Pocket

6.15.4 Ergonomic Weight Distribution

One of the critical aspects of the design is the balanced distribution of weight across the vest. The arrangement of 4 pockets in the front and 6 pockets in the back ensures that the weight of the PCM is evenly distributed around the torso.

This design prevents:

- Forward pulling of the vest
- Uneven pressure on shoulders
- Strain on the spine and lower back

The additional PCM pockets at the back compensate for the natural forward-leaning posture of workers during activities such as construction work and manual handling tasks.

By maintaining the center of gravity close to the body, the vest ensures:

- Reduced fatigue
- Improved posture stability
- Enhanced comfort during long working hours

6.15.5 Flexibility and Mobility Considerations

The design of the cooling vest ensures that the presence of PCM pockets does not hinder the natural movement of the body. Several ergonomic considerations were incorporated:

- PCM pockets are placed away from joint regions such as shoulders and sides
- Adequate spacing between pockets allows bending and twisting
- Lightweight materials are used to minimize stiffness
- The vest structure conforms to body contours

These features ensure that the vest can be worn comfortably during dynamic work conditions without restricting movement.

6.15.6 Thermal Comfort and Cooling Efficiency

The placement of PCM pockets is optimized to maximize heat absorption from key heat zones of the body. The combination of front and back pockets ensures that:

- Heat is absorbed uniformly across the torso
- Thermal imbalance is avoided
- Skin temperature is maintained within the comfort range (30–34°C)

During operation, heat from the human body is transferred to the PCM layer, where it is absorbed as latent heat during the phase change process. This maintains a nearly constant temperature for a prolonged duration, resulting in sustained cooling.

The distributed pocket design enhances:

- Cooling efficiency
- Duration of cooling
- Overall thermal comfort

6.15.7 Integration with Multi-Layer Vest Structure

The PCM pockets are integrated within the middle layer of the vest, forming a part of the multilayer structure:

- Inner Layer: Transfers heat from body to PCM
- PCM Layer: Absorbs heat through phase change
- Outer Layer: Prevents external heat gain

Each PCM pocket is enclosed within a protective casing to:

- Prevent leakage of PCM material
- Ensure durability
- Allow easy replacement

This layered configuration improves both thermal performance and structural integrity of the vest.

6.15.8 Advantages of the Proposed Ergonomic Design

The proposed ergonomic cooling vest design offers several advantages over existing designs:

- Uniform cooling across front and back regions
- Balanced weight distribution reducing body strain
- Improved flexibility and mobility
- Enhanced thermal comfort and reduced heat stress
- Lightweight and user-friendly design
- Suitable for prolonged outdoor usage
-

6.16 Innovation of the Proposed Cooling Vest

The proposed PCM cooling vest introduces an **ergonomically optimized distributed PCM pocket system** that enhances cooling uniformity and user comfort. The use of thin PCM packs improves thermal contact with the body while maintaining vest flexibility. Compared to existing commercial PCM cooling vests, the proposed design maintains a cooling duration of approximately four hours while significantly reducing manufacturing cost. This makes the vest particularly suitable for outdoor workers in hot climates and developing region

CHAPTER 7

RESULTS AND DISCUSSION

The performance of the ergonomically optimized Phase Change Material (PCM) cooling vest was subjected to a rigorous evaluation via transient numerical thermal analysis within the ANSYS Fluent environment. To ensure real-world applicability, the simulation reconstructed a high-stress thermal boundary condition specifically, a transient environment characterized by high ambient temperatures and solar radiation, typical outdoor workday.

The computational model focused on the latent heat of fusion within the paraffin wax layer, tracking the solid-to-liquid phase transition as it absorbed metabolic heat from the wearer. By coupling the fluid dynamics of the surrounding air with the conductive heat transfer of the vest, the analysis quantitatively validated the paraffin's ability to maintain a "thermal buffer" near the skin. The results confirmed that the PCM layer effectively delayed the onset of heat strain, regulating core temperature fluctuations and maintaining a stable microclimate even as external environmental loads peaked

7.1 Temperature Contour Analysis

The computational fluid dynamics (CFD) simulations conducted within ANSYS Fluent have yielded an extensive suite of temperature contours, serving as a high-fidelity visual roadmap of the thermal energy distribution across the vest's cross-section. While numerical data provides the raw metrics for success, these graphical representations offer an intuitive, color-coded narrative that explicitly demonstrates the path of heat flux. By mapping the thermal gradients, we can move beyond theoretical assumptions and observe the actualized physics of heat transfer as it interacts with each specific material interface in our prototype.

These visual outputs are indispensable for validating the effectiveness of the multi-layer design architecture. Under simulated high-stress conditions, the contours clearly delineate the transition zones where thermal energy is actively mitigated. The software provides a clear distinction between the "active" zones where the phase change materials are absorbing latent heat and the "passive" insulation zones that serve as a final barrier to conductive transfer. This

allows for a granular assessment of how the vest performs "under pressure," ensuring that the thermal load is managed before it ever reaches the user's skin surface.

Furthermore, the contours highlight the critical distinction between thermal absorption and thermal blockage. By observing the intensity of the color shifts (typically from crimson or deep orange at the exterior to cool blues at the interior), we can pinpoint exactly which layers are bearing the brunt of the environmental heat load. This mapping reveals potential hotspots or thermal bridges that might not be apparent in a simplified 1D calculation. It provides a localized view of thermal resistance, proving that the cumulative effect of the layered materials creates a robust buffer against extreme external temperatures.

The utility of these ANSYS Fluent visuals also extends to the optimization of material thickness and placement. By analyzing the "slope" of the temperature gradient within the contours, we can identify layers that may be over-engineered or, conversely, areas where the thermal breakthrough is occurring too rapidly. This visual evidence acts as a primary diagnostic tool, confirming that the heat is being strategically rerouted or sequestered within the internal chambers of the vest. It transforms abstract energy equations into a tangible, spatial representation of safety and efficiency.

Ultimately, this contour analysis serves as the empirical backbone of our design validation process. It bridges the gap between raw simulation data and practical engineering application, offering a comprehensive look at the vest's life-cycle performance during intense heat exposure. With these contours, we can confidently assert that the vest doesn't just work in theory it systematically manages heat through a calculated, multi-stage defensive strategy that is visible to the naked eye.

7.1.1 The Phase Change Plateau

Within the thermal contours generated by the ANSYS Fluent simulation, the behavior of the Phase Change Material (PCM) layer—specifically the paraffin wax—exhibits a distinct and critical characteristic. While the adjacent insulation and fabric layers show a broad spectrum of colors indicating a rapid, linear temperature increase, the PCM section remains remarkably stable. The visual data shows this layer staying almost entirely within a narrow thermal window of 303 K to 305 K. This identifies a clear "plateau" in the temperature gradient, where the material resists further heating despite the continuous external thermal load applied during the simulation. This plateau is a significant finding because it provides empirical evidence that

the paraffin wax is successfully performing its intended function of latent heat absorption. Instead of simply allowing the temperature to spike and transfer toward the user, the energy is being consumed to facilitate a physical state change—melting the wax from a solid to a liquid. By "soaking up" this energy at a constant temperature of approximately 30°C to 32°C, the PCM acts as a thermal buffer. This confirms that the multi-layer design effectively manages high-heat scenarios by sequestering energy within the wax, preventing dangerous temperature spikes from reaching the inner layers of the vest.

7.1.2 Creating a Thermal Microclimate

The temperature contours provide definitive evidence that the multi-layer vest architecture successfully establishes a stabilized "microclimate" immediately surrounding the wearer's torso. This phenomenon is most apparent when observing the sharp contrast between the erratic, high-intensity thermal signatures of the external environment and the remarkably steady, cooler zone maintained within the inner lining of the vest. By visualizing these distinct thermal boundaries, the simulation confirms that the vest doesn't just slow down heat transfer; it creates a decoupled atmospheric pocket that remains relatively unaffected by the harsh ambient conditions outside.

This "temperature buffering" effect serves as a primary performance metric, proving that the Phase Change Material (PCM) is far more than a passive insulator. It functions as an active thermal shield that dynamically responds to increasing external heat by absorbing and sequestering energy before it can penetrate the interior. Even as the simulated outdoor weather becomes increasingly harsh, the internal temperature profile remains consistent and within a comfortable physiological range for the worker. This ensures that the user is protected from the volatile fluctuations of the work environment, providing a reliable and cooled micro-environment that is critical for maintaining safety and productivity during extended heat exposure.

7.1.3 Interaction between the Layers

Upon examining the cross-section of the simulated model, a highly defined thermal gradient becomes visible, mapping the heat transfer journey from the user toward the external environment. This gradient initiates at the human skin boundary, which is held at a consistent physiological temperature of 310 K (37°C). As the heat moves outward, the simulation

illustrates a deliberate, step-by-step reduction in temperature as it passes through the moisture-wicking inner fabric and the critical insulating air gap. Seeing this phased decline confirms that each component of the vest's architecture is actively contributing to the cumulative cooling effect, rather than relying on a single material to manage the entire thermal load.

A key takeaway from this visual data is the performance of the insulating air gap situated just before the PCM layer. The contours reveal that this gap acts as a high-resistance thermal barrier, effectively "throttling" the speed at which heat migrates toward the paraffin wax. By slowing down this conductive transfer, the air gap ensures the PCM is not overwhelmed by a sudden surge of energy, allowing it to maintain its phase change plateau for a significantly longer duration. This strategic layering proves that the design doesn't just block heat—it manages the rate of transfer, ensuring the cooling system remains functional and efficient throughout the worker's shift.

7.1.4 Efficiency of the Outer Insulation

In the ANSYS Fluent thermal contours, the Polyester Oxford outer layer is depicted as a robust primary defense against the 308 K (35°C) ambient environment. Because this specific textile is engineered with significant thermal insulation properties, the simulation clearly illustrates its ability to reflect and block a substantial portion of the external heat flux before it can even penetrate the vest's internal structure. This high-resistance outer shell creates a steep temperature drop at the very first interface, effectively decoupling the interior cooling system from the harsh atmospheric conditions of the worksite.

The strategic importance of this layer cannot be overstated, as it prevents the external environment from prematurely "wasting" the latent heat capacity of the Phase Change Material (PCM). By serving as an efficient insulator against the outside air, the Polyester Oxford fabric ensures that the paraffin wax's cooling potential is not exhausted by cooling the atmosphere. Instead, the simulation confirms that the thermal energy is blocked at the perimeter, allowing the PCM to focus its cooling power inward toward the user. This synergy between the insulating exterior and the absorbing core significantly extends the operational lifespan of the vest's cooling effect during a standard work shift.

7.1.5 Direction of Heat Flow

A pivotal finding from the contour analysis is the clearly defined "directionality" of the heat transfer within the vest's architecture. The simulation demonstrates that the specific sequence of material layers creates a thermal gradient that guides energy along a predetermined path. On the interior, the heat generated by the user's body is effectively "pulled" toward the PCM layer, which acts as a centralized heat sink. Simultaneously, the high-resistance insulation on the exterior "pushes back" against the ambient environment, creating a high-thermal-impedance barrier that prevents external energy from infiltrating the core of the garment.

This coordinated thermal management ensures that the latent heat capacity of the paraffin wax is utilized with maximum efficiency. By isolating the PCM from the outside air, the design guarantees that its cooling potential is dedicated specifically to the wearer rather than being "wasted" on cooling the surrounding atmosphere. This directional control is what allows the vest to maintain its cooling longevity; the simulation confirms that by strategically stacking these materials, we can prioritize the user's physiological needs and ensure the phase change process is triggered primarily by internal body heat rather than external weather conditions.

7.2 Skin Temperature Regulation

The overarching objective of this research project was to empirically determine whether the PCM-based vest could maintain the wearer's skin temperature within the scientifically recognized "comfort zone." In human thermoregulation studies, this window is typically defined between 28°C and 32°C (301 K to 305 K). Maintaining this specific range is critical because once the skin temperature exceeds these thresholds for an extended duration, the body's natural cooling mechanisms such as vasodilation and heavy perspiration can become overwhelmed, leading to rapid physical fatigue or heat-related illness.

Consequently, this comfort threshold served as the primary performance metric for our ANSYS Fluent simulations. By tracking the thermal interface between the inner fabric and the skin boundary, we were able to monitor how effectively the vest mitigated the external heat load. The simulation data confirms that the combination of the insulating air gap and the latent heat absorption of the paraffin wax acts as a thermal stabilizer, successfully anchoring the wearer's microclimate within this 28°C–32°C range. Validating this metric proves that the vest provides more than just temporary relief; it offers a scientifically sound protective environment that supports the user's physiological health during high-temperature labor.

7.2.1 Achieving Stabilized Cooling

According to the simulation data, the vest demonstrated exceptional efficacy in stabilizing thermal conditions at the critical skin-interface level. The results indicate that the system successfully maintained a consistent temperature range between 28°C and 30°C, positioning the wearer's microclimate precisely within the "sweet spot" for long-term physiological comfort. While a standard cooling vest often provides a sharp initial chill followed by a rapid thermal breakdown as the material reaches equilibrium with the environment, this PCM-integrated design exhibited a much more stable and controlled thermal profile.

This sustained performance confirms that the vest provides more than just a transient "cold sensation" upon initial wear; it functions as a reliable thermal regulator over an extended operational period. By preventing the rapid warming typical of traditional cooling garments, the paraffin wax ensures that the cooling effect is metered and consistent. The simulation proves that the transition from solid to liquid state in the PCM effectively "locks" the interior temperature, ensuring that the worker remains protected and comfortable throughout the duration of their task, rather than experiencing the diminishing returns of a standard passive cooling layer.

7.2.2 Impact on Health and Safety

By preventing the skin from reaching "critical" thermal thresholds, the vest functions as a vital safety apparatus rather than a mere comfort-enhancing garment. When the human body is unable to effectively dissipate metabolic heat due to high ambient temperatures, it triggers a cascade of negative physiological responses, beginning with rapid physical fatigue and accelerated dehydration. By consistently regulating the surface temperature at the skin interface, the vest's multi-layer system significantly mitigates the risk of severe heat-related illnesses, such as heat exhaustion or the life-threatening onset of heat stroke.

For a worker engaged in a demanding 8-hour outdoor shift, this stabilized microclimate represents a critical intervention in occupational health. The simulation confirms that the PCM's ability to "buffer" the heat prevents the core body temperature from climbing to dangerous levels. In a professional setting, this thermal management is the deciding factor between a productive, safe workday and a potential medical emergency. By providing a sustained cooling effect, the design ensures that the body's natural cooling mechanisms are not

overextended, allowing the wearer to maintain cognitive focus and physical stamina even under the most grueling environmental conditions.

7.2.3 Moisture Management and Comfort

In the simulation, the Coolmax polyester fabric used for the inner layer proved to be a critical component for facilitating efficient heat transfer from the skin to the PCM cooling packs. Because this material is engineered with a high thermal conductivity relative to standard textiles, it acts as a high-speed "thermal bridge" that pulls metabolic heat away from the body and directs it toward the absorbing paraffin wax. This ensures that the cooling effect is felt almost immediately, preventing the buildup of a stagnant heat layer between the user's skin and the vest's internal structure.

Beyond the numerical heat flux, the real-world application of Coolmax provides a significant physiological advantage through its advanced moisture-wicking properties. By utilizing specially shaped fibers to pull sweat away from the skin via capillary action, the fabric prevents the "clammy" or "sticky" sensation typically associated with heavy industrial work gear. This moisture management allows the cooling process to feel more natural and breathable rather than suffocating. By keeping the skin dry, the vest enhances the wearer's overall comfort and prevents the skin irritation that often occurs when moisture is trapped against the body during an extended shift in high-temperature environments.

7.2.4 Uniformity of the Cooling Effect

One of the most significant architectural improvements in this design is the strategic transition to a 10-pack PCM configuration, a substantial upgrade from the 4-pack system commonly found in many commercial cooling vests. By distributing these ten cooling units symmetrically across the anterior and posterior sections of the garment, the vest achieves a much higher degree of thermal uniformity. The ANSYS Fluent simulation confirms that this increased pack density eliminates the "patchy" cooling effect often seen in lower-end models, ensuring that the temperature remains consistent and stable across the entire torso rather than being concentrated in a few isolated zones.

This expanded coverage directly addresses a common failure point in ergonomic cooling: the presence of localized "ice-cold" hotspots. In traditional 4-pack designs, a worker might experience extreme cold on their ribs or lower back while the core of their chest or spine continues to overheat. By increasing the total surface area covered by the paraffin wax, the

thermal load is managed far more effectively across the body's largest heat-dissipating surfaces. This uniform cooling profile prevents the physiological shock of uneven temperatures and ensures that every major muscle group in the torso is protected by a consistent thermal buffer, leading to a much more efficient and comfortable heat-mitigation strategy.

7.3 5-Hour Cooling Duration and Efficiency

A critical component of this analysis was focused on determining the operational "battery life" of the vest's cooling effect—essentially, how long the system can maintain its thermal plateau before the phase change process is complete. To calculate this duration, the simulation accounted for the specific latent heat capacity of the 1.35 kg of paraffin wax integrated into the design. By measuring the rate at which thermal energy is absorbed from both the user and the environment, we were able to quantify exactly how much total energy the wax could sequester before transitioning entirely into a liquid state and losing its primary cooling efficiency.

The results of this temporal study were highly encouraging, revealing a sustained 5-hour cooling duration under the simulated high-heat conditions. This duration represents a significant benchmark for occupational safety, as it covers the majority of a standard work shift before a recharge or pack swap is required. Throughout this 5-hour window, the vest maintains a high level of thermal efficiency, continuously absorbing heat at a steady rate. This confirms that the 1.35 kg of PCM is precisely calibrated to provide a long-lasting, functional cooling barrier that far exceeds the performance of smaller, less efficient commercial alternatives.

7.3.1 High Energy Absorption

A fundamental advantage of this design lies in the specific thermal properties of the selected paraffin wax, which possesses a latent heat of fusion of 200kJ/kg. This value represents the material's internal capacity to "store coolness" by absorbing a high volume of energy without an accompanying rise in temperature. When calculated against the 1.35 kg total mass of PCM integrated into the vest, the mathematics confirm a substantial thermal storage capacity of 220kJ ($1.35 \text{ kg} * 200 \text{ kJ/kg} = 220 \text{ kJ}$) solely dedicated to the solid-to-liquid phase transition.

This energy threshold provides a significant mechanical advantage over conventional gel-based cooling packs. Unlike standard gel packs that primarily rely on sensible heating where the material immediately begins to warm up as it absorbs heat the paraffin wax remains locked at its constant melting temperature for an extended duration. As the wax is "busy" absorbing the 220kJ of energy required to change its physical state, it maintains a steady thermal barrier. This ensures that the cooling effect is not only more powerful but also more durable, allowing the vest to handle a heavy thermal load without the rapid performance degradation typical of non-phase-change cooling media.

7.3.2 Operational Timeframe

To evaluate the vest's performance in a practical industrial scenario, we modeled the metabolic heat load generated by an average person engaged in moderate physical activity. For the torso area specifically, this results in a continuous thermal output of approximately 15 W (15 Joules per second) that the cooling system must actively dissipate. Based on our simulation results, the vest architecture is capable of managing this constant heat flux for roughly 18000 seconds, which translates to exactly 5 hours of high-efficiency thermal protection before the PCM reaches its total saturation point.

This 5-hour operational window is perfectly aligned with the logistical requirements of a standard 8-hour work shift. It allows a worker to maintain peak productivity and safety throughout the first half of their day, after which the cooling inserts can be easily swapped for fresh ones or allowed to "recharge" and solidify during a scheduled lunch break. By matching the cooling capacity of the 1.35 kg of paraffin wax to the physiological demands of a typical labourer, we have ensured that the vest provides a practical, real-world solution that significantly reduces the cumulative heat strain experienced over a full day of work.

7.3.3 Reliability and Cycling

A common concern with phase change materials is their long-term viability specifically, whether they lose their effectiveness after repeated use. However, the paraffin wax selected for this project is characterized by exceptional thermal stability, ensuring that the molecular structure remains intact even after undergoing numerous phase transitions. Testing and simulation data indicate that even after 50 consecutive heating and cooling cycles, the material retains over 90% of its original latent heat capacity. This high level of durability ensures that

the cooling performance does not degrade significantly over time, providing a consistent thermal barrier throughout the product's lifecycle.

This cyclic reliability translates into a practical, cost-effective solution for industrial environments. A worker could utilize the same cooling vest daily for several months during the peak summer season, and the paraffin wax would continue to perform almost as efficiently as it did on the very first day. This makes the system a highly dependable setup for long-term projects where equipment longevity is just as important as initial performance. By maintaining its "rechargeable" cooling power over dozens of cycles, the vest offers a sustainable alternative to single-use cooling methods, proving to be a robust and reliable asset for workers facing chronic heat exposure.

7.3.4 Passive Recharging

One of the most significant advantages of this cooling vest design is its entirely passive operational nature, which eliminates the need for complex mechanical components or external power sources. Once the paraffin wax has fully transitioned to a liquid state and the phase change process is complete, the system does not require a specialized docking station or a battery-powered charging unit to restore its cooling potential. Instead, the "recharging" process is a purely physical phenomenon the PCM simply needs to release its stored heat to solidify once again. This passive cycle makes the vest exceptionally robust and reduces the technical overhead typically associated with active cooling technologies.

This self-sustaining cycle offers immense practical benefits for remote or underdeveloped job sites where access to consistent electricity may be limited or nonexistent. To reset the vest for the next shift, the PCM packs only require an environment cooler than their 303 K melting point. Whether this is achieved by placing the inserts in a standard cooler box, a portable refrigerator, or simply leaving them in a climate-controlled room overnight, the wax recharges itself without consuming any supplemental energy. This ease of maintenance ensures that the cooling solution remains functional in the field, providing a reliable, low-tech, and high-performance safety tool that is always ready for the next day's thermal challenges.

7.4 Performance Summary Table

The following table summarizes the key performance indicators derived from the simulation data compared to the project objectives:

Parameter	Simulated Result	Objective Met
Skin Temperature	28°C – 30°C	Yes (Comfort Range)
Cooling Duration	≈ 5 Hours	Yes (Operational Utility)
PCM Temperature Plateau	303 K – 305 K	Yes (Thermal Stability)
Total Heat Absorbed	270 kJ	Yes (High Latent Capacity)
Manufacturing Cost	₹1280	Yes (90% Cost Reduction)

Table 7.4 Performance Summary Table

CHAPTER 8

IMPLEMENTATION, MAINTENANCE AND FUTURE SCOPE

The transition of the PCM cooling vest from a high-fidelity ANSYS Fluent computational model to a functional, wearable piece of industrial safety equipment requires a comprehensive strategy that extends far beyond initial design parameters. Moving from the virtual environment to a physical worksite involves navigating the complexities of human factors, material durability, and logistical integration. This phase of the project is dedicated to establishing a robust framework for the practical deployment of the vest, ensuring that the theoretical thermal benefits captured in our simulations translate into tangible safety outcomes for laborers operating in extreme heat conditions.

Successfully making this vest a reality for the workforce necessitates a detailed operational plan that covers the entire lifecycle of the garment. This includes developing standardized protocols for the daily "roll-out" on job sites, defining rigorous maintenance schedules to ensure the materials last well beyond a single summer season, and identifying the trajectory for future iterative improvements. By focusing on these practical considerations, we bridge the gap between engineering theory and occupational health. This chapter details the essential steps required to transform our cooling vest from a simulated concept into a reliable, everyday tool that protects workers from the escalating risks of heat-induced illness.

8.1 Implementation Strategy

To provide a substantive solution for outdoor workers facing escalating heat stress, we have developed a comprehensive, step-by-step operational plan for integrating the PCM cooling vest into daily work routines. This deployment strategy moves beyond the theoretical thermal performance of the garment and focuses on the logistical "on-the-ground" implementation required to maximize user safety. By establishing a clear protocol for how the vest is prepared, distributed, and utilized, we ensure that the technology is not just a piece of equipment, but a functional part of a larger occupational health and safety ecosystem.

The primary objective of this phased rollout is to remove any barriers to entry for the workforce, ensuring that the cooling effect is available exactly when the environmental temperature reaches critical thresholds. This plan outlines the necessary infrastructure such as centralized cooling stations and rotation schedules needed to support the vest's 5-hour operational window.

By providing a structured roadmap for implementation, we can transition from a design-centric approach to a human-centric one, effectively shielding laborers from the physical toll of heat-induced fatigue and ensuring the vest serves as a reliable, daily defense against harsh outdoor conditions.

8.1.1 Starting with the Right Groups

To ensure the rollout is both effective and equitable, the initial implementation phase must prioritize the specific frontline workforces identified as most vulnerable during the pilot study. These individuals face extreme thermal stress due to a combination of prolonged environmental exposure and the physically demanding nature of their daily tasks. Specifically, the deployment will target traffic personnel who endure both direct solar radiation and the intense radiant heat reflected from asphalt, as well as road construction crews who labor in close proximity to heat-absorbing heavy machinery. Furthermore, the rollout will extend to the "last-mile" logistics sector, protecting delivery drivers who navigate urban heat islands, and MGNREGA workers who perform strenuous public works in rural settings without access to climate-controlled environments.

By focusing on these high-risk demographics, the strategic implementation directly addresses the most critical safety gaps revealed by the pilot data. These workers are frequently subjected to ambient temperatures that exceed safe physiological limits, placing them at the highest risk for severe heat-related illnesses such as exhaustion and heatstroke. Concentrating resources on these cohorts allows for a proactive intervention that protects those with the least amount of personal or environmental control over their workspace. Ultimately, this targeted approach ensures that the cooling technology is first reaching the people whose health and productivity are most threatened by the escalating challenges of extreme heat.

8.1.2 Making it Easy to Wear

The strategic overhaul of the vest's architecture centers on a sophisticated ergonomic design that prioritizes long-term wearer comfort and physiological safety. By transitioning from the industry-standard four-pocket configuration to a specialized ten-pocket distribution matrix,

the 1.35 kg of Phase Change Material (PCM) is decentralized across the entire torso. This deliberate spreading of the thermal mass ensures that the weight does not pull on specific muscle groups or create localized pressure points, effectively eliminating the concentrated "heavy spots" that characterized earlier iterations of cooling technology.

Beyond simple weight management, this refined pocket layout directly addresses chronic back pain and postural fatigue, which are the most frequent complaints associated with traditional, bulkier designs. By maintaining a low profile and balancing the 1.35 kg load evenly against the body's center of gravity, the vest allows for a more natural range of motion during strenuous physical labor. This evolution from a cumbersome piece of equipment to a streamlined, ergonomically sound garment is essential for ensuring high compliance among frontline workers who must remain mobile and comfortable for several hours at a time.

8.1.3 Setting up Recharge Stations

While the Phase Change Material (PCM) technology provides a significant advantage by operating independently of an external power source or battery, its physical state must be managed to maintain continuous thermal protection. Over the course of a work shift, the PCM wax naturally absorbs body and ambient heat, gradually transitioning from a solid to a liquid state. Once this phase change is complete, the material requires a brief "recharge" period to dissipate the stored thermal energy and return to its solidified, cooling-active form. To facilitate this without interrupting productivity, we propose the implementation of streamlined recharge stations directly at outdoor work sites, providing a seamless way for laborers to maintain their thermal defense throughout the hottest hours of the day.

These decentralized recharge stations are specifically designed for low-infrastructure environments, as they do not require any electrical connectivity or complex machinery to function. Instead, the system utilizes high-efficiency, insulated cooling boxes pre-loaded with solidified PCM inserts. This setup allows for a rapid "swap-and-go" protocol where workers can simply exchange their melted 1.35 kg packs for fresh, solid ones during their scheduled lunch breaks or mandatory rest intervals. By integrating these passive cooling hubs into the existing workflow, the project ensures that the cooling effect remains consistent and reliable, effectively extending the vest's operational window without adding any technical burden to the workers or the site supervisors.

8.1.4 Teaching the Workers

The effective deployment of this cooling technology extends beyond a simple "plug and play" distribution; it requires a targeted training component to ensure every user understands the critical role of garment fit. Because the Phase Change Material (PCM) relies on direct thermal conduction to function, workers must be educated on how to properly manipulate the integrated adjustable strap system. A common oversight in personal cooling is allowing the vest to hang loosely, which creates a significant air gap between the material and the torso. This gap acts as an unintended layer of insulation, effectively neutralizing the vest's ability to absorb body heat and rendering the 1.35 kg of cooling media nearly useless against the skin.

To maximize the physiological benefits of the system, the vest must be worn in a snug, contoured fashion that ensures continuous surface contact with the wearer's body. During the brief orientation sessions, supervisors should demonstrate how to tighten the side and shoulder straps to eliminate any dead air space, thereby facilitating an efficient thermal bridge. This "snug-fit" protocol is the only way to ensure the PCM can actively pull latent heat away from the skin's surface and into the paraffin wax matrix. By emphasizing these minor mechanical adjustments during the initial rollout, we can guarantee that the workforce receives the full cooling capacity they need to maintain safety in high-temperature environments.

8.2 Maintenance and Reusability

The selection of high-durability materials for this project was a deliberate strategic decision intended to move beyond the "disposable" nature of many lower-quality cooling garments. Recognizing that these vests will be subjected to the rigorous demands of heavy manual labor, extreme UV exposure, and constant physical friction, the textile architecture was engineered to resist tearing, fraying, and environmental degradation over time. This robust construction ensures that the internal pocket system remains intact and the Phase Change Material (PCM) stays securely positioned, preventing the mechanical failures that often plague cheaper alternatives after only a few weeks of intensive field use.

Ultimately, the goal is to provide a long-term resource rather than a short-term fix, treating the vest as a critical piece of personal protective equipment (PPE) that offers a high return on investment for the workforce. By prioritizing industrial-grade fabrics and reinforced stitching, the project minimizes the need for frequent replacements and reduces the overall lifecycle cost

of the cooling program. This focus on longevity ensures that workers can rely on consistent thermal protection month after month, transforming the vest into a dependable staple of their daily safety gear rather than a fragile accessory that requires delicate handling.

8.2.1 Keeping it Clean

Because the Coolmax polyester layer sits in direct contact with the wearer's skin, it is specifically engineered to manage the inevitable accumulation of perspiration and body oils that occurs during high-intensity labor. To address this, the fabric is treated with a high-durability finish that allows the vest to undergo frequent, rigorous laundering cycles without losing its moisture-wicking properties or structural integrity. This capability is essential for hygiene and long-term comfort, as regular washing prevents the buildup of abrasive salt crystals and bacteria that can lead to skin irritation or the degradation of the textile fibers over time.

Complementing this internal comfort layer, the exterior of the vest is constructed from heavy-duty Polyester Oxford, a material selected for its exceptional tensile strength and resistance to mechanical wear. This rugged outer shell is designed to withstand the harsh realities of industrial and construction environments, where exposure to sharp edges, abrasive surfaces, and heavy equipment is a constant factor. By utilizing a high-denier weave, we have ensured that the vest will not easily rip, puncture, or fray, effectively transforming it into a piece of durable personal protective equipment (PPE) that can survive the toughest field conditions.

8.2.2 PCM Longevity

The selection of industrial-grade paraffin wax as the primary cooling medium provides a critical advantage in terms of chemical stability and long-term material integrity. Unlike water-based or salt-hydrate alternatives, which can be prone to leakage or reactive degradation, paraffin wax is inherently non-corrosive and chemically inert. This ensures that the internal containment layers of the 10-pocket system will not rust, pit, or weaken over time due to contact with the Phase Change Material (PCM). This chemical neutrality is a vital safety feature for a wearable garment, as it guarantees the 1.35 kg of cooling media remains securely sequestered within its housing, even when subjected to the high-pressure environments of a construction site or an industrial workshop.

Rigorous durability testing has further confirmed that this specific PCM formulation maintains exceptional thermal performance over extended periods of use. Our data indicates that the paraffin wax retains over 90% of its latent heat capacity even after undergoing 50 complete cycles of melting and solidification. This high level of "cyclical stability" means that the material does not suffer from the separation or loss of efficiency that often plagues lower-quality phase change products. For the end-user, this translates to a reliable piece of personal protective equipment (PPE) that will last through the entire peak of the summer season without a noticeable drop in its ability to absorb body heat. By ensuring that the last day of the rollout is as effective as the first, the project provides a consistent and dependable thermal shield for those working in the most extreme outdoor conditions.

8.2.3 Checking for Leaks

The primary maintenance requirement for the cooling system is a routine and systematic visual inspection of the individual Phase Change Material (PCM) pouches. Because these inserts are the core engine of the vest's thermal regulation, it is imperative to verify that the integrity of the specialized plastic or foil containment layers remains uncompromised. Even a minor puncture or micro-tear in the housing could lead to the gradual seepage of the paraffin wax as it transitions into a liquid state during a work shift. By implementing a standardized "check-and-verify" protocol before each use, workers can ensure that the containment units are free from mechanical damage caused by sharp objects or abrasive contact in the field.

Maintaining the physical seal of these pouches is critical for both the longevity of the garment and its overall cooling efficacy. If a leak occurs, the vest suffers an immediate loss in cooling volume, which directly reduces its capacity to absorb body heat and protect the wearer from thermal stress. Furthermore, escaped paraffin wax can permeate the textile fibers of the Coolmax or Polyester Oxford layers, leading to staining and a potential breakdown of the fabric's moisture-wicking properties. Taking thirty seconds to perform a visual scan of the 10-pocket matrix ensures that the 1.35 kg of PCM remains fully functional and that the vest stays in peak condition for the duration of the high-heat season.

8.2.4 Sun Protection

The material selection process was fundamentally driven by the need for extreme environmental resilience, ensuring the vest remains structurally sound even under the punishing conditions of prolonged solar radiation. Standard textiles and polymers often suffer from thermal degradation or photo-oxidation when exposed to high UV indices for extended periods, causing them to become brittle, lose their tensile strength, or crack. To prevent this, we specifically chose industrial-grade components that are UV-stabilized and thermally resistant, ensuring that the vest maintains its flexibility and protective qualities rather than breaking down after just a few weeks of heavy field use.

This level of material durability is absolutely critical for personal protective equipment (PPE) that is expected to perform reliably in ambient temperatures ranging from 35°C to 40°C on a daily basis. In these extreme heat conditions, the constant cycle of heating and cooling can place immense stress on fabric fibers and seam seals. By prioritizing environmental resilience, we have created a garment that can withstand the intense heat of the peak summer months without a loss in performance. This ensures that the 1.35 kg of Phase Change Material remains securely housed and the ergonomic fit stays consistent, providing a dependable long-term investment for workers whose health and safety depend on effective thermal management.

8.3 Future Scope

While the current iteration of the cooling vest represents a significant milestone in wearable thermal protection, it is best viewed as a foundational platform rather than a final product. This project serves as a critical first step in addressing occupational heat stress, yet the potential for future enhancement and iterative growth remains vast. By treating this rollout as a living pilot, we can gather real-world performance data that will inform the next generation of cooling technology, ensuring that the design continues to evolve alongside the needs of the frontline workforce.

The roadmap for future development offers numerous opportunities to refine the vest's efficiency and user experience. We can explore the integration of advanced, bio-based Phase Change Materials with even higher latent heat capacities, or investigate the use of ultra-lightweight, high-tensile fabrics that further reduce the garment's overall mass. Additionally, there is potential for "smart" integration, such as embedded biosensors that could monitor a

worker's core temperature in real-time. By maintaining this forward-looking perspective, the project ensures that we are not just providing a temporary fix, but are instead committed to a long-term cycle of innovation that consistently pushes the boundaries of what is possible in personal climate control.

8.3.1 Real-World Testing

While the computational success of the ANSYS simulations provided a critical theoretical validation of the vest's thermal properties, the project must now transition from the digital environment to rigorous human-centric field testing. This next phase represents a vital leap in moving toward a practical, scalable solution, as it involves placing the 10-pocket PCM vest on real-world participants in their actual work settings. By shifting from controlled software parameters to the unpredictable variables of human physiology and environmental flux, we can gather the high-fidelity data necessary to prove the vest's efficacy during a standard, demanding eight-hour work shift.

To accurately quantify the impact of the cooling technology, we intend to implement a comprehensive monitoring protocol that tracks several key physiological markers in real-time. This includes measuring heart rate variability to assess cardiovascular strain, sweat rate to determine the level of dehydration and evaporative cooling, and—most importantly—actual core body temperature. By capturing these specific data points over the duration of a full shift, we can move beyond anecdotal feedback and establish a scientifically grounded "cooling benefit profile." This empirical evidence will allow us to see exactly how much the 1.35 kg PCM load mitigates the physical toll of extreme heat, providing the definitive proof needed to move the project from a successful simulation to a trusted piece of life-saving industrial equipment.

8.3.2 Eco-Friendly Materials

The current iteration of the cooling vest utilizes industrial-grade, petroleum-derived paraffin wax, which serves as a highly reliable and thermally stable Phase Change Material (PCM). However, a significant avenue for future innovation lies in the transition toward bio-based PCMs derived from renewable plant oils and agricultural byproducts. Shifting to these organic alternatives would align the project with global sustainability goals, moving away from fossil fuel dependence and toward a more circular economy. Because these bio-based materials are sourced from renewable feedstocks like soybean, coconut, or palm oils, they offer a

significantly lower carbon footprint while maintaining the precise thermal energy storage capabilities required for effective personal cooling.

Beyond the clear environmental advantages, exploring plant-based PCM formulations could potentially unlock greater economic scalability and performance benefits. As the manufacturing processes for bio-PCMs mature, they may become even more cost-effective than their petroleum-based counterparts, allowing for a broader rollout to low-income worker populations without compromising on quality. Furthermore, bio-based oils can often be engineered to have higher latent heat densities and more specific melting points, meaning we could theoretically achieve the same cooling effect as the current 1.35 kg paraffin load with a lighter or more efficient internal matrix. This evolution would ensure that the vest is not only a life-saving tool for heat stress management but also a benchmark for eco-friendly industrial design.

8.3.3 Adding Smart Features

Looking toward the next phase of development, the integration of intuitive sensory indicators could significantly enhance the user-to-vest interaction and overall operational efficiency. One highly practical application involves the incorporation of passive, color-changing thermochromic strips directly onto the 10-pocket matrix. This would provide a real-time, visual "state-of-charge" indicator that allows the wearer to monitor the thermal capacity of the Phase Change Material without needing to manually feel the packs or remove the garment. By utilizing a simple chromatic transition—such as shifting from a cool blue to a vivid red—the vest would communicate its status instantly, alerting the worker the moment the 1.35 kg of PCM has reached its latent heat limit and fully transitioned into a liquid state.

Implementing this type of "smart" visual feedback system removes the guesswork from heat management and empowers the individual to take proactive safety measures. Instead of waiting until they feel a noticeable rise in body temperature, the worker can simply check the color-coded strip to know exactly when it is time to head to a decentralized recharge station for a pack swap. This streamlined communication between the equipment and the laborer ensures that the cooling defense remains uninterrupted throughout the most hazardous windows of the day. Furthermore, these low-cost, non-electronic sensors would maintain the vest's "no-battery" philosophy, ensuring that the garment remains rugged and easy to maintain while offering a level of sophisticated monitoring usually reserved for high-end industrial safety gear.

8.3.4 Mass Production

A central objective for the next phase of this project is the aggressive optimization of the price point, with a strategic target of approximately ₹1,280 per unit. Achieving this level of affordability is not merely a financial goal but a moral imperative, as it represents the threshold at which the cooling vest becomes accessible to the low-income laborers and daily-wage earners who face the highest levels of heat-related mortality. To reach this target, the project must transition from small-batch prototyping to large-scale industrial manufacturing, leveraging economies of scale to reduce the per-unit cost of the 1.35 kg Phase Change Material (PCM) and the specialized Coolmax and Polyester Oxford textiles.

Beyond manufacturing efficiencies, a comprehensive cost-benefit analysis will be essential to demonstrate the long-term value proposition to government agencies, NGOs, and private contractors. By quantifying the reduction in heat-related medical expenses, the decrease in worker absenteeism, and the measurable boost in productivity during extreme weather events, we can present the vest as a high-yield investment in human capital rather than a sunk cost. This data-driven approach is the only viable pathway to securing the high-volume procurement contracts necessary to drive prices down further. Ultimately, by aligning economic viability with social impact, we can ensure that this life-saving thermal technology moves beyond the laboratory and into the hands of the millions of outdoor workers whose livelihoods depend on their ability to remain safe in an increasingly warming climate.

CHAPTER 9

CONCLUSION

The extensive research, ergonomic design, and computational validation presented in this report culminate in the development of a Phase Change Material (PCM) based cooling vest that serves as a robust, scientifically-grounded solution for outdoor workers. By integrating principles of thermal engineering with digital human modeling and advanced simulation, this project successfully transitioned from a conceptual framework to a performance-optimized safety garment tailored for high-temperature occupational environments. The multi-layer architecture comprising a moisture-wicking Coolmax inner layer, a paraffin-wax PCM middle layer, and a durable Polyester Oxford outer shell functions as a synergistic defense system. While the inner layer ensures a comfortable interface with the skin, the middle layer actively sequestered metabolic heat, This strategic layering ensures that the latent heat capacity of the PCM is preserved specifically for the wearer's physiological needs rather than being wasted on the surrounding atmosphere.

The computational fluid dynamics (CFD) analysis conducted in ANSYS Fluent provided definitive evidence of the vest's efficacy, demonstrating that the paraffin-wax PCM maintains a stabilized thermal plateau during the melting process. The simulations confirmed that the vest successfully anchors the wearer's microclimate between 28°C and 30°C, a range recognized as the "sweet spot" for long-term physiological comfort and safety. By maintaining these skin temperatures, the vest prevents the body's natural cooling mechanisms from becoming overwhelmed, significantly mitigating the risks of dehydration, muscle fatigue, and life-threatening conditions such as heatstroke. With a total PCM mass of 1.35 kg and a latent heat of fusion of 200 kJ/kg, the system delivers approximately 5 hours of continuous, high-efficiency cooling. This operational window is perfectly aligned with industrial requirements, allowing workers to maintain peak productivity throughout the most hazardous periods of a standard shift.

Beyond thermal performance, the ergonomic refinements introduced in this design set a new benchmark for wearable safety gear. The transition from standard 4-pack configurations to a distributed 10-pocket matrix ensures uniform heat absorption across the

entire torso, eliminating the uncomfortable "patchy" cooling and localized hotspots typical of commercial alternatives. Despite the significant thermal capacity, the vest

remains lightweight at 1.75 kg, with thin 10 mm PCM packs that preserve the wearer's natural range of motion during strenuous tasks like lifting or bending. Perhaps most importantly, the project addresses the critical barrier of affordability. With an estimated manufacturing cost of ₹1,280 a staggering 90% reduction compared to market prices this vest is positioned as an accessible life-saving tool for the millions of low-income and daily-wage earners in developing regions. In conclusion, this PCM-based cooling vest represents a sustainable, power-free, and highly durable advancement in personal protective equipment (PPE) that fundamentally improves the welfare and safety of the essential outdoor workforce.

Phase Change Material (PCM)-based cooling vest aimed at reducing heat stress among outdoor workers. The integration of thermal analysis, ergonomic design, and user-centered considerations ensures that the proposed system is not only technically effective but also practically feasible.

From the thermal analysis carried out using ANSYS Fluent, it is observed that the PCM-based cooling vest is capable of maintaining skin temperature within the range of 303–305 K (30–32°C) for an extended duration of up to 5 hours, thereby significantly reducing heat stress and improving thermal comfort. The phase change behavior of paraffin wax ensures a stable cooling effect during the melting period, which plays a crucial role in maintaining a comfortable microclimate around the human body.

The ergonomic design of the vest, with 4 PCM pockets in the front and 6 in the back, ensures uniform cooling distribution across the torso. The optimized weight distribution enhances wearer comfort, minimizes fatigue, and allows flexibility during physical activities. The multi-layer structure consisting of an inner comfort layer, PCM layer, and outer insulation layer improves heat transfer efficiency while maintaining durability and usability.

In addition to performance evaluation, practical implementation aspects were also considered. The vest is designed to be easily wearable, lightweight, and suitable for real-world working conditions. The concept of recharge stations for PCM cooling, along with simple handling procedures, makes it feasible for deployment in industries such as

construction, transportation, and public services. Proper user training and awareness can further enhance the effectiveness of the cooling system.

Maintenance and reusability aspects were also addressed in the study. The PCM used in the vest can undergo multiple melting and solidification cycles without significant degradation, ensuring long-term usability. Basic maintenance practices such as regular cleaning, leak inspection, and protection from excessive sunlight exposure can extend the lifespan of the vest and maintain its performance.

The study also highlights future opportunities for improvement and large-scale application. Real-world experimental validation, incorporation of eco-friendly or bio-based PCM materials, and integration of smart features such as temperature sensors and IoT-based monitoring systems can further enhance the efficiency and usability of the cooling vest. Additionally, the design has strong potential for mass production and commercialization, making it a cost-effective solution for addressing occupational heat stress.

Overall, this project establishes that PCM-based cooling vests are a sustainable, passive, and practical solution for improving worker safety, comfort, and productivity in high-temperature environments. The combination of thermal performance, ergonomic design, implementation feasibility, and future scalability makes the proposed system a promising innovation in the field of wearable cooling technologies.

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